West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Computer Science and Technology [CST], Computer Science and Engineering [CSE], Computer Software Technology [CSWT] & Information Technology [IT]

Part-II (3rd Semester)

Revised 2022

Detailed Syllabus for 3rd Semester

Computer Science and Technology, Computer Science and Engineering, Computer Software Technology & Information Technology

Sl. No	Category	Code No.	Course Title	Hours per week			Total contact hrs/ week	Credits
				L	T	P		
1.	Program core course	CST201	Computer Programming	2	0	0	2	2
2.	Program core course	CST203	Scripting Languages (Python)	2	0	0	2	2
3.	Program core course	CST205	Data Structures	2	0	0	2	2
4.	Program core course	CST207	Computer System Organization	3	1	0	4	4
5.	Program core course	CST209	Algorithms	3	1	0	4	4
6.	Summer Internship-I (4 weeks) after 2nd Sem	SI201	Summer Intern-ship-1					2
7.	Program core course	COPC211	Computer Programming Lab	0	0	4	4	2
8.	Program core course	CST213	Scripting Languages Lab	0	0	4	4	2
9.	Program core course	CST215	Data Structures Lab	0	0	2	2	1
	Total Credits							21

Course Title: Computer Programming in C	
Course Code	CST201
Number of Credits :2	2 (L: 2, T: 0, P: 0)
Prerequisites	Ability to develop logic / flow of simple problem.
Course Category	PC
Course code: CST	**Semester: THIRD**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs/week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

- **To study the structure programming concept.**
- **To study Linear Data Structure.**
- **To study Looping and Branching.**
- **To study subscripted variables and user defined data types.**
- **To study user defined functions.**
- **To study pointers in depth.**

Course Objectives

To enable student, develop structured solutions to problems and implementing them using computers. This involves two parts:

- **Formulating a solution for a given problem as a well-defined sequence of actions, and**
- **Expressing solution in a machine-readable form or a programming language.**

For the second part, we will learn the common units of programming languages. The first part can only be learned through the repeated practice of solving problems.

Course Content:

Contents (Theory)	Hrs./Unit	Module
UNIT 1: Basics of C	5	A

- History of C, Advantages of Structured Program, Files (source, header, object, binary executable) used in C, Characteristics of C.
- C character set, Tokens, Constants, Variables, Keywords, Data types used in C.
- C operators (arithmetic, logical, assignment, relational, unary, binary, increment and decrement, conditional, ternary, bit wise, special, comma, sizeof, postfix, prefix etc.), Operator precedence, Associativity of operators, Type conversion, Typecasting.
- Formatted input, Formatted output.

UNIT 2: Decision Control and Looping Statements	**5**	A

- Decision making and branching statements, if statement (if, if-else, else-if ladder, nested if-else), Switch case statement & applications.
- Conditional and unconditional ‘goto’ statement and drawbacks.
- Iterative/Loop statement, Entry controlled & exit controlled loop structure & differences, Example like while, do- while, and for loop structure, Break and continue statement & their uses, nested loop structure & applications.

UNIT 3: Subscripted Variables / Arrays	10	B

- Advantages of subscripted variables/ arrays & accessing array elements, Declaration and initialization of one dimensional, two dimensional, multidimensional (idea only) and character arrays & Strings, Accessing array elements.
- Declaration and initialization of string variables, String handling functions from standard library (strlen(), strcpy(), strcat(), strcmp()), Applications like string operations to extract substring from left, right, middle of a string, Replacement of string characters, concatenation of two strings etc.

UNIT 4: User defined functions	10	C

- Definition of functions, advantages of functions in modular approach problem solving, Prototype declaration, Scope and lifetime of variables & Storage Class (Auto, Extern, Static, Register), defining functions, function signature, passing parameter types, Function call (call by value, call by reference), Return values.
- Recursion and use of memory stack, Types of recursion. Recursion vs Iteration. Applications.

UNIT 5: Pointers in C	15	C

- Understanding pointers, difference between memory variables and pointer variables, Declaring and accessing pointers, constant pointers and pointer to a constant, Null Pointers, Generic Pointers, Pointers arithmetic and expressions.
- Passing arguments to function using pointers, Pointers and arrays, Passing an array to a function, Array name and Pointer.
- Pointers and Strings, Array of pointers, Function pointers, Pointer to a pointer.
- Dynamic memory allocation using malloc(), calloc(), realloc(). Uses of free(). Pointer to a structure.

Course outcomes

- **Student should be able to computationally formulate basic problems and write code to execute them.**
- **The focus of the course as mentioned above should be on example-based learning.**

Reference Books

1. Programming in C, Author: Reema Thareja, OXFORD University Press
2. Computer Fundamentals and Programming In C, Author Anita Goel & Ajay Mittal, Pearson
3. Head first C, Author by Griffiths, SPD publication
4. C: Concepts & Programming, Author: Bhusry, Willy
5. Programming in C, Author: E. Balagurusamy, TMH
6. Mastering C, Author: Venugopal, TMH
7. C programming: Test your skills, Author: Kamthane Ashok, Pearson Education India

Course Title : Scripting Languages	
Course Code	CST203
Number of Credits :2	2(L: 2, T: 0, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : THIRD
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 2 hrs/week	Continuous Internal Assessment : 20 Marks
Tutorial: - 0	Attendance-10 Marks Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Total Contact Hoirs:30 Hours	
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Learn Scripting Language features and programming

Course Objectives:

- **To learn how to work with a scripting language.**
- **To introduce Python programming language through its core language basics and program design techniques suitable for modern applications.**
- **To understand the wide range of programming facilities available in Python covering graphics, GUI, data visualization.**
- **To utilize high-performance programming constructs available in Python to strengthen applications and development in practical scenarios.**

Course Content:

Contents (Theory)	Hrs./Unit	Module
UNIT 1: Introduction, Variables and Data Types	**7**	**A**

- **History, Features, setting up path, Installation and Working with Python, Basic Syntax**
- **Understanding Python variables**
- **Numeric data types: int, Long int, float, complex/imaginary**
- **String data type and string operations: String literals, manipulating strings, comparing strings, Unicode string literals, converting between Simple Types, Converting to strings, String Formatting, String Methods.**
- **Date and Time**
- **Other Data Types:**
  - **Tuples**
  - **List: Defining list, list slicing, Split, Join, Manipulating Lists, Copying Lists**
  - **Dictionary**
  - **Arrays**
- **Basic Operators: Arithmetic Operators, Relational Operators, Assignment Operators (Simple assignment and Multiple Assignment-Tuple packing & unpacking) Logical Operators, Bitwise Operators, Membership Operators, Identity Operators**

- **Operator Precedence**
- **Understanding coding blocks**

UNIT 2: Control Structures	**5**	**A**

- **Conditional blocks using if, else and elif**
- **For loops and iterations**
- **while loops**
- **Loop manipulation using continue, break and else and pass**
- **Programming using conditional and loops block**
- **Modify loops : break and continue**

UNIT 3: Functions, Modules and Packages	**6**	**B**

Organizing codes using functions

- **Defining Functions & Calling Functions**
- **Pass by object reference**
- **Parameters**
- **Arbitrary arguments**
- **Optional and Named Arguments**
- **Passing arguments from a tuple**
- **Variable Scope and Binding: Local Variables, Nonlocal Variables, Global Variables, class scope**

Organizing projects into modules: Grouping Code with Modules
Importing own module as well as external modules
Understanding Packages: Grouping Modules into Packages

UNIT 4: : File I/O, Text Processing, Regular Expressions	**6**	**C**

Accessing Keyboard Input: raw_input and input
Printing to the Screen: print
File modes and permissions
read functions: read(), readline(), readlines()
write functions :write(),writelines()
other file operations: open(),close(),tell(),seek(),flush(),fileno(), isatty(),next()
Redirecting output streams to files.
Programming using file operations
Powerful pattern matching and searching: re.match(),re.searh(),re.findall(), re.finditer()
Creating and Using Regular Expression Objects- import re, re.compile(),re.sub(), re.subn(), re.split()
Power of pattern searching using regex

UNIT 5: **Frameworks**	**6**	**C**

Frameworks - The MVC framework, Django- What is Django and why should you use Django?

Creating URL, Templates

• Send data to a template
• Display data in a template
• Display object lists in a template

- **Handle chains with filters in Django**
- **Use URLs effectively**
- **Create base templates in order to extend other templates**
- **Insert static files in our templates**

Django Form

- **Create an HTML form**
- **Handle the data sent by a form**
- **Create a Django form**
- **Validate and manipulate data sent from a Django form**
- **Create forms based on models**
- **Customize error messages and usage of widget**

Reference Books

1. **Taming Python by Programming, Jeeva Jose, Khanna Publishing House**
2. **Core Python Programming, Dr. R. Nageswara Rao, Dreamtech Press**
3. **Python Programming: Using Problem Solving Approach, Reema Thareja, Oxford University Press**
4. **Core Python Programming, Wesley J. Chun, Pearson**
5. **Python for programming, P. Deitel, H. Deitel, Pearson**
6. **Starting Out with Python, Tony Gaddis, Pearson**
7. **Introduction to Computation and Programming Using Python. John V. Guttag, MIT Press.**
8. **An Introduction to Python, G.v. Rossum, SPD**
9. **Practical Programming: An Introduction to Computer Science using Python 3, Paul Gries, The Pragmatic Bookshelf**

Course outcomes: **At the end of the course student will be able to build program with a scripting language and will be able to learn any other scripting language on their own.**
Design real life situational problems and think creatively about solutions of them. Apply a solution clearly and accurately in a program using Python. Apply the best features of Python to program real life problems.

Course Title: Data Structures	
Course Code	CST205
Number of Credits :2	2 (L: 3, T: 0, P: 0)
Prerequisites	Basic Knowledge of Computer system
Course Category	Computer Science and Technology
Course code: CST	**Semester: THIRD**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs/week **Total Contact Hours: 45 Hours**	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

- **To study the structure programming concept.**
- **To study Linear Data Structure.**
- **To study Dynamic Memory Allocation.**
- **To study Non-Linear Data Structure.**
- **To study user defined Graf theory.**
- **To study Recursion in depth.**

Course Objectives

To enable student, develop structured solutions to problems and implementing them using computers. This involves two parts:

- **Formulating a solution for a given problem as a well-defined sequence of actions, and**
- **Expressing solution in a machine-readable form or a programming language.**

To provide strong foundation for implementing programming language to formulate, analyze and develop solutions related to various data structures problems.

Course Content:

Contents (Theory)	Hrs./Unit	Module
UNIT 1: Introduction to Data Structures	3	A

- ✓ **Basic Terminology**
- ✓ **Classification of Data Structures**
- ✓ **Operations on Data Structures.**

UNIT 2: Linear Data Structures	**12**	A

- ✓ **Stacks:**
  - **Introduction to Stacks**
  - **Array Representation of Stacks**
  - **Operations on a Stack**
  - **Applications of Stacks-Infix-to-Postfix Transformation**
  - **evaluating Postfix Expressions.**
- ✓ **Queues:**
  - **Introduction to Queues**
  - **Array Representation of Queues**
  - **Operations on a Queue**
  - **Types of Queues-De-Queue**
  - **Circular Queue**
  - **Applications of Queues-Round Robin Algorithm.**
- ✓ **Recursion (GCD, Tower of Hanoi Problem)**

UNIT 3: Linked Lists	15	B

- ✓ **Singly Linked List**
  - **Representation in Memory,**
  - **Operations on a Single Linked (add new node- first, in-between, end position. Delete-- first, in-between, end position),**
- ✓ **Circular Linked Lists**
  - **Operations on a circular Single Linked (add new node- first, in-between, end position.**
  - **Delete-- first, in-between, end position),**
- ✓ **Doubly Linked Lists**

- **Operations on a Double Linked (add new node- first, in-between, end position. Delete-- first, in-between, end position),**

✓ **Circular Linked Lists**

- **Operations on a circular Double Linked (add new node- first, in-between, end position.**
- **Delete-- first, in-between, end position),**

✓ **Linked List Representation**

- **Operations of Stack,**
- **Operations of Queue.**

UNIT 4: Non-Linear Data Structures	15	C

✓ **Trees:**

- **Basic Terminologies**
- **Definition and Concepts of Binary Trees**
- **Representations of a Binary Tree using Arrays and Linked Lists**
- **Operations on a Binary Tree-Insertion, Deletion**
- **Traversals, Types of Binary Trees**
- **B-Tree**
- **AVL Tree**

✓ **GRAPHS:**

- **Graph Terminologies**
- **Representation of Graphs- Set, Linked**
- **Matrix**
- **Graph Traversals**
- **BFS and DFS**

Course outcomes

- **Student should be able to computationally formulate basic problems and write code to execute them and have a good understanding of Data Structures and its applications in algorithms**
- **The focus of the course as mentioned above should be on example-based learning.**

Reference Books

1. Introduction to Data Structures in C, Kamthane, Pearson
2. Data Structures Using C, Reema Thareja, Oxford University Press India.
3. Data Structures, Lipschutz Seymour, McGraw-Hill Education
4. A simplified approach to data structures, Pawan Goyal, Published by SPD
5. Data Structures Using C, 1e, Tenenbaum, Pearson
6. Data Structures and Algorithms, Aho, Pearson

Course Title : Computer System Organization	
Course Code	CST207
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	Basic Knowledge of Computer system and Digital gates
Course Category	Computer Science
Course code : CST	Semester : THIRD
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Theory : - 3 hrs/week, Tutorial: - 1 hr/week
Examination Scheme	Continuous Internal Assessment : 20 Marks
	Attendance-10 Marks Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Total Contact Hoirs:60 Hours	
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop the concept of how computer works internally.

Course Learning Objectives: To have a thorough understanding of the basic structure and operation of a digital computer, it's Architecture, computational designs and how computer works.

Course Content:

Contents (Theory)	Hrs./Unit	Module
UNIT 1:	6	A

Structure of Computers:

- **Computer Functional units, Von-Neumann architecture, Bus structures, Basic Operational Concepts, Data representation (Fixed and Floating point), Error detecting codes.**

Register Transfer and Micro Operations:

- **Register transfer, Memory transfers, Arithmetic micro-operations, Logic micro-operations, Shift micro-operations, and Arithmetic logic shift unit.**

UNIT 2:	**20**	A

Micro Programmed Control:

- **Control memory, Address sequencing, and design of control unit.**

Computer Arithmetic:

- **Addition and Subtraction, Multiplication and Division algorithms, Float- ing-point arithmetic operation,**

Pipelining:

- **Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline**
- **Vector Processing, Array Processors.**

UNIT 3:	**10**	B

Introduction to Microprocessor Architecture:

- **Instruction Set, Architecture design principles from programmer's perspective.**
- **One example microprocessor (Intel 8086), Block diagram, Pin functions, Register structure, Segmentation, Interrupt mechanism, Addressing modes, Instructions.**

UNIT 4:	**9**	B

Assembly Language Programming:

- **Simple programs, Assembly language programs involving logical, branch and call instructions,**
- **sorting, evaluation of arithmetic expressions, string manipulation,**
- **assembler directives, procedures and macros.**

UNIT 5:	**15**	C

Memory and Digital Interfacing:

- **addressing and address decoding, interfacing RAM, ROM, EPROM,**
- **programmable peripheral interface,**
- **Cache Memory (Mapping and Hit ratio),Virtual Memory Technique (Logical address, Physical address, TLB etc.)**

Reference Books

8. **Computer Organization, Carl Hamacher, Zvonks Vranesic, SafeaZaky, McGraw-Hill**
9. **Computer Organization and architecture, William Stallings, Pearson**
10. **Microprocessors Interface, Douglas V.Hall, Tata McGraw-Hill.**
11. **Microprocessors and Microcontrollers, Senthil Kumar, Saravanan, Jeevananthan, Oxford**
12. **Computer System Architecture, M. Moris Mano, Pearson/PHI, India.**
13. **The X86 Microprocessors - Architecture and Programming and Interfacing, Das, Pearson**
14. Computer organization design and Architecture 5th edition, Shiva (SPD/CRC press)
15. **Advanced Microprocessors and Peripherals- Architecture, Programming and interfacing,A.K.Ray, K.M.Bhurchandi, Tata McGraw-Hill, New Delhi, India.**
16. **Computer Organization and Design: A Hardwar/Software Interface (MIPS Edition) by Patterson and Hennessy**

Course outcomes:

- **Have a good understanding of functioning of computer system as such and its various subcomponents.**
- **Student will be able to understand computing requirement for a specific purpose,**
- **Analyseperformance bottlenecks of the computing device and choose appropriate computing device fora given use case.**
- **Acquire Knowledge to write assembly language programs**

Course Title : Algorithms	
Course Code	CST209
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : THIRD
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 3 hrs/week	Continuous Internal Assessment : 20 Marks
Tutorial: - 1 hr/week	Attendance-10 Marks Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Total Contact Hoirs:60 Hours	
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop basic concept of Algorithms in Computer Science

Course Objectives: The objective of this course is to prepare the student with the algorithmic foundations of computing. A sound grasp of algorithms is essential for any computer science engineer. Almost all programming involves algorithms at some level.

Course Content:

Contents (Theory)	Hrs./Unit	Module
UNIT 1: Fundamentals of Algorithms	6	A

- **Definitions and Characteristics of Algorithm. Examples.**
- **Data Abstraction.**
- **Sets, Multisets,**
- **Stacks, Queues.**
- **Asymptotic Notations (*Order Notation*, Omega Notation, Theta Notation) with Examples.**
- **Time and Space Complexity. Best Average and worst-case analysis of algorithms.**
- **Programming Models Concepts: -Divide and Conquer, Greedy Methods, Dynamic Programming,**

UNIT 2: Sorting	**16**	A

The sorting problem.

- **Bubble sort.**
- **Selection sort.**
- **Insertion sort.**

- **Shell sort**
- **Merge sort.**
- **Quicksort.**
- **Heapsort.**

Computation of Best Average and worst-case Time complexity of all the above sorting algorithms.

Linear Time sorting

- **Count Sort**
- **Bucket Sort**
- **Radix Sort**

UNIT 3: Searching	**16**	B

Linear Search Algorithm.
Binary Search Algorithm.
Computation of Best, Average and Worst-case Time complexity of Linear and Binary Search

Binary Search Trees: Algorithms, Searching Time & space complexity.

Balanced Search Trees: What is the signification and advantage of height balancing?
Insertion, Deletion and Searching Algorithms of different types of Balanced Search Trees and their comparative study.

Hashing, Hash Tables Hash functions, Collision and Collision resolving techniques.

Symbol Tables

UNIT 4:	**16**	C

Definitions: Graph Directed and Undirected graph-Examples.
Paths, Cycles, Spanning trees-Examples.
Directed Acyclic Graphs-Examples.

Topological Sorting.

Minimum Spanning Tree algorithms:

- **Prim's Algorithm with Examples**
- **Kruskal's Algorithm with Examples**

Shortest Path algorithms:

- ✓ **Dijkstra's algorithm.**
- ✓ **Bellman–Ford Algorithm,**

Floyd-Warshall all pairs shortest path algorithm

UNIT 5: **Strings**	**6**	C

- **String Sort.**
- **Tries.**
- **Search a Substring within a string.**

- **String Matching Algorithms and their complexity analysis**
  - ✓ **Simple/Naive String Matching Algorithm**
  - ✓ **Rabin-Karp Algorithm**
  - ✓ **Knuth-Moris-Pratt Algorithm**
  - ✓ **Horspool String Matching Algorithm**
  - ✓ **Boyer-Moore String Matching Algorithm**
- **Regular Expressions.**
- **Elementary Data compression.**

Reference Books

1. **Algorithm Design, Jon Kleinberg| Eva Tardos, Pearson**
2. **Introduction to Algorithms, Cormen, Leiserson, Rivest and Stein. MIT Press**
3. **Design and Analysis of Algorithms, S.Sridhar, Oxford University Press.**
4. **Fundamentals of Computer Algorithms, E. Horowitz, S. Sahani, S. Rajasekaran, Galgotia.**
5. **Introduction to the Design and Analysis of Algorithms, Anany Levitin, Pearson**
6. **Algorithms in a Nutshell, G.T.Heinemam, SPD**
7. **Design & Analysis of Algorithms, Gajendra Sharma, Khanna Publishing House**

Course outcomes:

- **The student should be able to understand the basic notions of time and space complexity of algorithms.**
- **The student should be able to design basic algorithms for sorting and searching.**
- **The student should be able to realize Graph concepts, Minimum spanning Tree algorithms and shortest path algorithms.**
- **The student should be able to implement sorting, searching, tree and graph algorithms in a modern computer programming language**
- **The student should be able to understand String sort and how to search a substring within a string.**

Examination Scheme of ESE (End Semester Examination)				
	Question Type	**Question to be set**	**Questions to be answered**	**Marks**
Theoretical	**MCQ type questions carrying 1 mark.**	**15**	**10**	**10**
	Fill in the blanks type questions carrying 1 mark.	**15**	**10**	**10**
	Short answer type questions carrying 1 mark.	**15**	**10**	**10**
	Subjective type questions carrying 2 marks.	**10**	**6**	**12**
	Subjective type questions carrying 6 marks.	**9 (3 each from each of 3 modules)**	**3**	**18**
TOTAL				**60**

Course Title: Computer Programming Lab in C	
Course Code	CST211
Number of Credits :2	3 (L: 0, T: 0, P: 3)
Prerequisites	Basic Operations on Computer
Course Category	PC
Course Code: CST	**Semester: THIRD**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Laboratory: 3 hrs/week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 60Marks External Assessment: 40 Marks

Course Objectives:

This Lab course is intended to practice what is taught in theory class of 'Computer Programming' and become proficient in computer programming. Computer programming is all about regular practice. Students should work on solved and unsolved problems listed in the text books, and the problems given by the teacher. Some of the topics that should necessary be covered in lab are listed below.

Course Content:

Sr. No.	Topics for Practice	
01	Familiarization with programming environment (Editor, Compiler, etc.)	Skill Area
02	a) Displaying hexadecimal, decimal, octal number format of the entered numbers. b) Displaying entered number with leading zeros and trailing zeros. c) Displaying entered number with right and left justification. d) Displaying with different formatting specifiers.	Programs using Formatted input and output.
03	a) To find greatest / smallest of three numbers. b) To display pass class, second-class, distinction according to the marks entered from the keyboard. c) To find even or odd numbers. d) To display spellings of number 1-10 on entry. e) Implementation and displaying the menu to execute 1. ADD, 2. SUBTRACT 3. MULTIPLICATION, 4. DIVISION using switch case. f) To check whether there exist real (float) roots of a quadratic equation and if exist find them.	Programs using decision making statements and branching statements
04	a) To display our College name twenty times on screen. b) To demonstrate Continue and Break statements within loop structure. c) To add first 'n' natural, even, odd numbers using different loop structures. d) To find GCD, LCM of two integral numbers. e) To generate simple number triangle for n rows. f) To generate Pascal triangle for n rows. g) To add the series 1 + (1 + 2) + (1 + 2 + 3) + …+ (1 + 2 + 3 + … + n) h) To generate all prime numbers within the given range.	Programs using loop statements

	i) To find all the Armstrong numbers within 100 to 1000.	
05	a) To find the largest and smallest numbers from array elements. b) To sort array elements in ascending / descending order. c) To enter elements for 3X3 matrix and display them. d) To calculate addition / subtraction of 2-dimensional matrix. e) To calculate multiplication of 2-dimensional matrix. f) To find the number of vowels and consonants in a string. g) Implementation of strlen(), strcpy(), strcat() and strcmp() functions. h) To check whether a string is palindrome or not. i) To replace a specific character/string by another character/string in a multiword string. j) To make the abbreviated form of a multiword string.	Programs to demonstrate applications of 1 & multi-dimensional arrays & Strings
06	a) To calculate the value of ${}^{n}C_{r}$, $n \geq r$ using function b) To find the sum of the series $1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}$ for $n \geq 1$, $x \geq 0$ using function. c) To interchange the biggest and smallest number in to calculate factorial a one-dimensional array using function. d) To calculate addition, subtraction and multiplication of 2-dimensional matrix using function. e) Write a program in C to find GCD of two numbers using recursion. f) To calculate factorial of any given number using recursion. g) To demonstrate call by reference, call by value.	Programs to demonstrate parameter passing mechanism & recursion.
07	a) To read and display an integer array using pointer. b) To read and display a text using a character pointer to a string. Also count the number of characters, words and lines in the text. c) To read, display, add and subtract of two times defined using hour, minutes and values of seconds. d) To read and display the contents of a structure variable using pointer to a structure. e) Write a program in C to create a singly linked list of n nodes and display it in reverse order. f) Write a program in C to insert a new node to a Singly Linked List after a desired node and display the list. g) Write a program in C to delete a node from a Singly Linked List after/ before a desired node and display the list. h) Implement Stack and Queue data structure using dynamic memory allocation.	Programs to demonstrate use of pointers and dynamic memory allocation.

Course outcomes

- Use of programming language constructs in program implementation.
- Apply different logics to solve given problem.
- Write program using different implementations for the same problem.
- Identify different types of errors as syntax, semantic, fatal, linker & logical.
- Debugging of programs.
- Understanding different steps and stages to develop complex program.

Reference Books

1. Programming in C, Author: Reema Thareja, OXFORD University Press
2. Programming in C, Author: E. Balagurusamy, TMH
3. C in Depth, Author: Srivastava, BPB
4. Mastering C, Author: Venugopal, TMH
5. C programming: Test your skills, Author: Kamthane Ashok, Pearson Education India
6. Head First C, David Griffiths, SPD

Course Title: Scripting Languages Lab	
Course Code	CST213
Number of Credits :2	2(L: 0, T: 0, P: 4)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : THIRD
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	**Examination Scheme** **1.Continuous Internal Assessment :60 Marks** This 60 Marks will be comprised of the following Marks division: • **Assignment: 20 Marks** • **Class Performance: 20 Marks** • **Viva Voce: 10 Marks** • **Attendance: 10 Marks** **2.End Semester Examination (External Assessment/ Sessional) :40 Marks**
Practical: - 4 hrs/week	

Course Objectives: This Lab course is intended to practice whatever is taught in theory class of Scripting Languages' and become proficient in scripting. Computer programming is all about regular practice. Students should work on solved and unsolved problems listed in the text books, and the problems given by the teacher. Some of the topics that should necessary be covered in lab are listed below.

Course Content:

Contents (Practical)	
Installing Python in Windows/Linux/Ubantu/Mac OS	
Sr. No.	**Topics for Practice**

S.No.	Topics for Practice
1	Practice basic coding syntax
2	Write and execute scripts based on data types
3	Write and execute Python scripts with conditionals and loops
4	Write and execute Scripts based on Functions and Modules
5	File Processing scripts
6	Write and execute Regular Expressions
7	Develop a simple web application

Some of the suggested sample programs are:

- **Running instructions in Interactive interpreter and a Python Script**
- **Write a script to purposefully raise Indentation Error and Correct it**
- **Write a script to find Sum and average of first n natural numbers**
- **Given 2 strings, s1 and s2, create a new string by appending s2 in the middle of s1**
- **Write a script to check whether a given string is palindrome or not.**
- **Write a program add.py that takes 2 numbers as command line arguments and prints its sum.**
- **Write a script using a for loop that loops over a sequence**
- **Write a script to count the numbers of characters in the string and store them in a dictionary data structure**
- **Write a program to use split and join methods in the string and trace a birthday with a dictionary data structure.**
- **Write a script that combines more than one lists into a dictionary**
- **Compute the GCD & LCM of two numbers.**
- **Check a number is prime or not**
- **Find the square root of a number**
- **Exponentiation (power of a number)**
- **Find all primes within a given range**
- **Find First n Fibonacci numbers.**
- **Find the maximum of a list of numbers**
- **Linear search and Binary search**
- **Write a function nearly equal to test whether two strings are nearly equal. Two strings a and b are nearly equal when a can be generated by a single mutation on b.**
- **Find the most frequent words in a text read from a file**
- **Programs that take command line arguments (word count)**
- **Write a function to find all duplicates in the list**
- **Remove empty strings from the list of strings**
- **Write a program to print each line of a file in reverse order.**
- **Write a program to compute the number of characters, words and lines in a file.**
- **Write a script to Calculate age in year month days of a person taking his/her date of birth as input and accessing current system date.**
- **Write a regular expression to search digit inside a string**

Finding Whether PIP is installed, Install virtual environment using pip, Installing Django, Setting PATH , Creating and Deploying Applications Using Django

Some of the suggested sample Exercises using Django

- **Send data to a template**
- **Display data in a template**
- **Display object lists in a template**
- **Handle chains with filters in Django**
- **Create and Use URLs in Django**
- **Create base templates in order to extend other templates**
- **Insert static files in our templates**
- **Create an HTML form**
- **Handle the data sent by a form**
- **Create a Django form**
- **Validate and manipulate data sent from a Django form**
- **Create forms based on models**
- **Customize error messages and usage of widget**

Course outcomes

Course outcomes: At the end of the course student will be able to build program with a scripting language and will be able to learn any other scripting language on their own.

Reference Books

1. **Taming Python by Programming, Jeeva Jose, Khanna Publishing House**
2. **Python Programming: Using Problem Solving Approach, Reema Thareja, Oxford University**
3. **Starting Out with Python, Tony Gaddis, Pearson**
4. **Core Python Programming, Wesley J. Chun, Prentice Hall**
5. **Introduction to Computation and Programming Using Python. John V. Guttag, MIT Press.**
6. **Beginning Python using Python 2.6 and Python 3, James Payne, Wrox publishing**
7. **Practical Programming: An Introduction to Computer Science using Python 3, Paul Gries, The Pragmatic Bookshelf**

Course Title: Data Structures Lab

Course Code	CST215
Number of Credits :2	2 (L: 0, T: 0, P: 4)
Prerequisites	Basic Knowledge of Computer system
Course Category	Computer Science and Technology
Course Code: CST	**Semester: THIRD**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Laboratory: 4 hrs/week Total Contact Hours: 60 Hours	Continuous Internal Assessment: 40 Marks Attendance, Assignment & Quiz: - 20 Marks End Semester Examination: 40 Marks

Course Objectives:

This Lab course is intended to practice what is taught in theory class of 'data Structure' and become proficient in computer programming. Computer programming is all about regular practice. Students should work on solved and unsolved problems listed in the text books, and the problems given by the teacher. Some of the topics that should necessary be covered in lab are listed below.

Course Content:

01	**Skills to be developed**	
	➢ Intellectual Skills: ➢ Use of programming language constructs in program implementation. ➢ To be able to apply different logics to solve given problem. ➢ To be able to write program using different implementations for the same problem ➢ Study different types of errors as syntax semantic, fatal, linker & logical ➢ Debugging of programs ➢ Understanding different steps to develop program such as ➢ Problem definition ➢ Analysis ➢ Design of logic ➢ Coding ➢ Testing ➢ Maintenance (Modifications, error corrections, making changes etc.)	Programs using Formatted input and output.
02	g) Motor Skills: Proper handling of Computer System.	

List of Practical:

01	Programs based on: Array operations, insertion, deletion
02	Programs based on Stacks Implementation of PUSH & POP operations, Evaluate postfix expressions, Infix to postfix h) conversions. Programs to demonstrate parameter passing mechanism & recursion.
03	Recursive programs: factorial, Fibonacci, Ackerman function, and tower of Hanoi.(any two)
04	Programs for demonstrating queue operations. one recursive program converted to non-recursive ones
05	Programs based on Linked lists
06	Programs based on trees Creating a binary tree, in order, pre order and post order traversal of binary tree, deleting a node from binary tree.
07	Programs for implementing various sorting techniques. (Minimum three sorting techniques from topics mentioned in the syllabus))
08	Programs for implementing various sorting and searching techniques. (Minimum two searching techniques from topics mentioned in the syllabus.)
09	Assignments based on graph theory.
10	Assignments based on graph theory.

LIST OF SAMPLE PROBLEMS FOR DATA STRUCTURE LAB (Example)

1. To write a program to check whether a word is palindrome or not.
2. To create a two-dimensional array of numbers and calculate & display the row & column sum and thegrand total.
3. To write a program of matrix multiplication.
4. To write a program to insert (Push) an element into the sack and delete (Pop) an element from the stackusing pointer.
5. To write a program to convert an infix expression to a postfix expression.
6. To evaluate a postfix expression.
7. To write a program to insert an element in the queue and delete an element from the queue using pointer.
8. To create a circular queue and add an element and delete an element from a circular queue.
9. To write a program of a structure containing an item name along with the unit price. The user enters the item name and quantity to be purchased. Program print outs total price of item with name using pointer ina structure or array in a structure.
10. To create a single linked list and — (a) insert a node in the list (before header node, in between two nodes, end of the list); (b0 delete a node from the list (1st node, last node, in between two nodes); (c) Concatenate two lists.
11. To create a doubly linked list and — (a) insert a node in the list (before header node, in between two nodes, end of the list); (b) delete a node from the list (1st node, last node, in between two nodes); (c) Concatenate two lists.
12. To create a circular linked list and insert & delete an element from the list.
13. Write a program to merge two shorted linked list.
14. Write a program to reverse a linked list.
15. To write a program to calculate the binomial co-efficient of ${}_nC^r$ of two numbers using recursive function.
    Also write the same program using function in non-recursive way.
16. To write a program to generate Fibonacci Series using recursive function. Also write the same programusing function in non-recursive way.
17. To write a program to create a binary tree and traverse it in pre-order and post-order form.
18. To write a program to create a binary search tree and —

    (a) insert a new node in the BST

    (b) search anode in the BST

    (c) delete a node from the BST

Course outcomes

- Using of programming language implement data Structure.
- Apply different logics to solve given problem.
- Write program using different implementations for the same problem.
- Identify different types of errors as syntax, semantic, fatal, linker & logical.
- Debugging of programs.
- Understanding different steps and stages to develop complex program.

Reference Books

1. Introduction to Data Structures in C, Kamthane, Pearson
2. Data Structures Using C, Reema Thareja, Oxford University Press India.
3. A simplified approach to data structures, Pawan Goyal, Published by SPD

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Computer Science and Technology [CST], Computer Science and Engineering [CSE], Computer Software Technology [CSWT] & Information Technology [IT]

Part-II (4th Semester)

Revised 2022

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Semester IV

CURRICULUM STRUCTURE FOR PART-II (SEMESTER 4) OF THE FULL-TIME DIPLOMA COURSE in Computer Science and Technology [CST], Computer Science and Engineering [CSE], Computer Software Technology [CSWT] & Information Technology [IT]

Sl. No	Category	Code No.	Course Title	Hours per week			Total contact hrs/ week	Credits
				L	T	P		
1.	Program core course	COPC202	Operating Systems	2	1	0	3	3
2.	Program core course	COPC204	Introduction to DBMS	2	1	0	3	3
3.	Program core course	COPC206	Computer Net-works	2	1	0	3	3
4.	Program core course	COPC208	SSAD/Software Engineering	3	0	0	3	3
5.	Program core course	COPC210	Object Oriented Programming Using Java.	3	0	0	3	3
7.	Minor Project	Proj.202	Minor Project	0	0	4	4	2
8.	Program core course	COPC212	Operating Systems Lab	0	0	2	2	1
9.	Program core course	COPC214	Introduction to DBMS Lab	0	0	2	2	1
10.	Program core course	COPC216	Computer Net-works Lab	0	0	2	2	1
11.	Program core course	COPC218	Object Oriented Programming Lab using Java	0	0	2	2	1
	Total Credits							**21**

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Course Title: OPERATING SYSTEMS	
Course Code	COPC 202
Number of Credits :3	3 (L: 2, T: 1, P: 0)
Prerequisites	Basic knowledge of computer organization, Data structure andhow computer works
Course Category	PC
Course code: CST	**Semester: FOURTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs/week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc: 10 MarksEnd Semester Examination: 60 Marks

Course Objectives

- Students will learn how Operating System is Important for Computer System.
- To make aware of different types of Operating System and their services.
- To learn different process scheduling algorithms and synchronization techniques to achieve better performance of a computer system.
- To know virtual memory concepts.
- To learn secondary memory management.

Course Content:

Contents (Theory)	Hrs./Unit	Marks
UNIT 1: Introduction to Operating System	5	8

- Overview of Operating System (OS),
- OS generations
- Operating system structure,
- OS Kernel,
- OS services and systems calls,
- System programs.

UNIT 2: Process Management and Concurrency Control	**20**	18

- Process Concept, Process states, Process control, Threads,
- Uni-processor Scheduling: Preemptive, Non preemptive, Scheduling algorithms:FCFS, SJF, RR, Priority, Thread Scheduling, Real Time Scheduling.
- IPC: Concept, Shared-Memory and Message-Passing Systems.
- Concurrency: Principles of Concurrency, Mutual Exclusion: S/W approaches, H/W Support, Semaphores, pipes, Message Passing, signals, Monitors,
- Classical Problems of Synchronization: Readers-Writers, and Dining Philosopher problem.
- Deadlock: Principles of deadlock, Deadlock Prevention, Deadlock Avoidance, Deadlock detection.

UNIT 3: Memory management	10	18

- Memory Management requirements, Memory partitioning: Fixed and Variable Partitioning,Memory Allocation: Allocation Strategies (First Fit, Best Fit, and Worst Fit), Fragmentation, Swapping, and Paging. Segmentation, Demand paging
- Virtual Memory: Concepts, management of VM, Page Replacement Policies (FIFO, LRU,Optimal, Other Strategies), Thrashing.

UNIT 4: I/O management & Disk scheduling:	6	10

- I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering.
- Disk Scheduling (FCFS, SCAN, C-SCAN, SSTF), RAID, Disk Cache.

UNIT 5: OS Security	4	6

- OS Security: Goals of Protection, Principle of protection, Access matrix, Access Control,Access Rights.
- Program Threats, Network and system threats, Cryptography, Authentication.

Course outcomes

- Understands the different services provided by Operating System at different level.
- They learn real life applications of Operating System in every field.
- Understands the use of different process scheduling algorithm andsynchronization techniques to avoid deadlock.
- They will learn different memory management techniques like paging, segmentation and demand paging etc.

Reference Books

1. Operating System Concepts, Silberschatz and Galvin, Wiley India Limited
2. Operating Systems, S. Haldar and Alex A. Aravind. Pearson Education, India
3. Operating Systems, Internals and Design Principles, Stallings, Pearson Education, India
4. Operating Systems, Nutt,Chaki and Neogy, Pearson Education, India
5. Modern Operating Systems, Andrew S. Tanenbaum, Prentice Hall of India
6. Operating systems, Deitel & Deitel, Pearson Education, India
7. Operating Syatems, Sumitradevi and Banashree, SPD

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Introduction to Operating System	A	3	3	2	8
2.	Process Management and Concurrency Control	A	6	6	6	18
3.	Memory management	B	6	6	6	18
4.	I/O management & Disk scheduling	C	3	4	3	10
5.	OS Security	C	2	2	2	6
	Total		**20**	**18**	**22**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels (Bloom's revisedtaxonomy)

Course Title: Introduction to DBMS	
Course Code	COPC 204
Number of Credits : 3	
Prerequisites	Having fundamental knowledge of Computers and elementary mathematics.
Course Category	PC
Course code: CST	Semester: Fourth
Duration: 15 weeks	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 3 hrs/week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

- To study the Structure of Database Management System.
- To study different data Modeling.
- To study Relational Data modeling, Algebra & Calculus.
- To study SQL.
- To study functional Dependency and Normalization techniques.

Course Objectives

To enable student, develop database driven applications using the capabilities provided by modern database management system software which involves

- ✓ Conceptual modeling.
- ✓ Relational database design
- ✓ And Database Query Languages.

Course Content:

Contents (Theory)	Hrs./Unit	Marks
UNIT 1: Database System Concepts & Architecture	6	5
1.1 Basic concepts of DBMS, Advantages of a DBMS over file processing system, Data Abstraction, Database Languages, Data Independence. 1.2 Components of a DBMS and overall ANSI / SPARC architecture of DBMS. 1.3 Roles and functionalities of DBA & Data Dictionary.		
UNIT 2: Data Modeling Technique	5	10
2.1. Data Models: Hierarchical Model, Network Model, Object Oriented Model, E-R Model. Comparative study between different models. 2.2. The Enhanced Entity-Relationship (EER) model: Specialization, Aggregation, Generalization.		
UNIT 3: Relational Data Model in Depth	12	15

3.1. Relational Model: Basic concepts, attributes and domains.
3.2. Keys Concept: Super, Candidate, Primary and Foreign key.
3.3. Integrity constraints: Implicit and Explicit Constraints. Domain, Key, Entity and Referential Integrity Constraints.
3.4. Security & Authorization: Secrecy, Integrity, Availability.
3.5. Relational Algebra and Calculus.

UNIT 4: Structured Query Language (should be taught preferably in Laboratory)	12	15

4.1. Introduction to SQL queries: Creating, Inserting, Updating and deleting tables.
4.2. Table creation applying constraints (Primary key, Check constraints etc.
4.3. Set operations & operators, Aggregate functions, string functions and date, time functions.
4.4. Null values, Nested sub queries, Complex queries, Join concepts.
4.5. PL/SQL Introduction: PL/SQL block structure, variables, SQL statements in PL/SQL, PL/SQL control Structures, Cursors, Triggers, Functions, Packages, procedures.

UNIT 5: Functional Dependencies & Normalization	10	15

5.1. Purpose of Normalization, Data redundancy and updating anomalies, Functional Dependencies and Decomposition.
5.2. Process of Normalization using 1NF, 2NF, 3NF, multivalued dependencies and BCNF.

Course outcomes

Student should be able to

- design a database, database-based applications
- understand the critical role of database system in designing several information system-based software systems or applications.
- use a DBMS in commercial query-based applications.

Reference Books

Name of Authors	Title of the Book	Edition	Name of the publisher
Abraham Silberschatz, Henry F. Korth, S. Sudarshan	Database System Concept	5th	TMH
Evan Bayross	Database concepts and Systems	3rd	SPD
P.S. Gill	Database Management System	2nd	Wiley
Date, Kannan & Swaminathan	An Introduction to Database Systems		Pearson
Navathe	Fundamentals of Database System		Pearson
Singh	Database Systems		Pearson

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Database System Concepts & Architecture	A	2	3		5
2.	Data Modeling Technique	A		4	6	10
3.	Relational Data Model in Depth	A	6	3	6	15
4.	Structured Query Language (should be taught preferably in Laboratory)	B	3	6	6	15
5.	Functional Dependencies & Normalization	C	4	6	5	15
	Total		**15**	**22**	**23**	**60**

Course Title : **Computer Networks**	
Course Code	COPC206
Number of Credits :	3 (L: 2, T: 1, P: 0)
Prerequisites	Having fundamental knowledge of Computers andelementary mathematics.
Course Category	CST
Course code : General	Semester : Fourth
Duration : 15 weeks	Total Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 2 hrs/week	Continuous Internal Assessment: 20 Marks
Tutorial: - 1 hr/week	Attendance, Assignment & Quiz : - 10, 5, 5 Marks respectively
Total Contact Hours: 45 Hours	End Semester Examination : 60 Marks

Aim:
One of the major components of computer based information systems is computer networks. Through computer networks we can share hardware, Software, Processing, Data and Applications besides getting global connectivity for internet based communication and services.
For diploma students it is important to understand the function of computer networks and obtain requisite knowledge about hardware and software requirements of networks and acquire skills to establish a network using necessary hardware & software tools and configure various services over it. The objectives of this course are to make students learn the technology of establishing, commissioning(making operational) and maintaining computer networks.

Course Objectives: Understand functioning of computer networks and popular networking protocols

Course Content:

Contents (Theory)	Hrs./Unit	Marks
UNIT 1: Basics of ComputerNetwork	4	4
1.1 Definition & history of networks 1.2 Usage of Computer Networks 1.3 Standard Organizations and Protocols 1.4 Line Configuration 1.5 Network Topology 1.6 Periodic Analog Signals (Sine Wave, Phase, Wavelength, Time and Frequency Domains, Composite Signals, Bandwidth) 1.6 Categories of network Based on scope Based on Connection 1.7 Performance (Bandwidth, Throughput, Latency (Delay), Bandwidth-Delay Product, Jitter)		
UNIT 2: The Reference Model for network communication	**3**	6

2.1 OSI model & function of each Layer 2.2 TCP/ IP model 2.3 Connection oriented v/s Connectionless approach Comparison of OSI & TCP/IP Models		
UNIT 3: Transmission Media	**5**	4
3.1 Types of Transmission Media 3.2 Guided Media: Twisted Pair, CoaxialCable, Fiber 3.3 Un Guided Media: Electromagneticspectrum, Radio Transmission, Microwave Transmission, Infrared Transmission, Satellite Communication.		
Unit 4: Multiplexing and Switching	4	**6**
4.1 Multiplexing (Frequency- Division, Web length- Division, Time Division) 4.2 Introduction of Switching (Three Methods of Switching, Switching and TCP/IP Layers)		
UNIT 5: Data Link Layer	**7**	8
5.1 Design issues, example protocols (Ethernet, WLAN, Bluetooth) 5.2 Link- Layer Addressing (Three type of Address, Address Resolution Protocol(ARP)) 5.3 Error Detection and Correction (Block Coding, Cyclic Codes, Checksum, Forward Error Correction) 5.4 Data Link protocols (Simple protocols, Stop-and-wait Protocol, Piggybacking) 5.6 Media Access Control (Random Access, Controlled Access, Channelization)		
Unit 6: Network Layer	**6**	8
6.1 Network Layer services (Packetizing, Routing and Forwarding) 6.2 Network-Layer Performance (Delay, Throughput, packet Loss, Congestion Control) 6.3 IPV4 Address (Address Space, Classful and Classless addressing) 6.4 Routing Algorithms		
Unit 7: Transport Layer	**7**	8
7.1 Transport layer Services 7.2 Connectionless and connection-oriented Protocols 7.3 Transport layer Protocols (Simple Protocol, Stop-and-Wait Protocol, Go-Back-N Protocol, Selective-Repeat Protocols, Bidirectional protocol: piggybacking)		
Unit 8: Network devices	**4**	8
8.1 Repeaters 8.2 Hubs		

8.3 Switches
8.4 Routers
8.5 Access Points
8.6 Gateways
8.7 Bridges
8.8 Difference between Layer 2 andLayer 3 Switches.
8.9 Introduction of Network Management software

UNIT 9: Application Layer	**5**	8

9.1 Principles of Network Application
9.2 The Web and HTTP
9.3 File Transfer: FTP
9.4 Electronic Mail in the Internet (SMTP, Comparison with HTTP, Mail Message Format, Mail access Protocols)
9.5 DNS
9.6 Peer-to-Peer Applications (P2P File Distribution)

Reference Book

1. Computer Networking by Kurose, Ross, Pearson;
2. Data Communications and Networking by Forouzan, Mc Graw Hill ;
3. Computer Networks by Andrew S Tannebaum, Pearson;
4. Head First Networking, SPD;
5. Computer Networks, Bhushan Trivedi, Oxford University Press
6. Networks for Computer Scientists and Engineers, Youlu Zheng & Shakil Akhtar, Oxford University Press
7. Data & Computer Communication by Williams Stallings, Pearson;

Course outcomes:

1. Understanding of computer networks, issues, limitations, options available.
2. Understanding of the care that needs to be taken while developing applications designed towork over computer networks
3. Able to configure basic LAN and connect computers to it.

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total Marks
I	Basics of Computer Network	A	4			4
II	The Reference Model for network communication	A	2	4		6
III	Transmission Media	A	4			4
IV	Multiplexing and Switching	A	2	4		6
V	Data Link Layer	B		2	6	8
vi	Network Layer	B	1	1	6	8
vii	Transport Layer	C	1	1	6	8
viii	Network devices	C	2	6		8
ix	Application Layer	C		6	2	8
	Total	**3**	**16**	**24**	**20**	**60**

Legends: R = Remembrance; U= Understanding; A= Application and above levels (Revised Bloom's taxonomy)

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Course Title : SOFTWARE ENGINEERING	
Course Code	**COPC 208**
Number of Credits :3	3 (L: 3, T: 0, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : FOURTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 3 hrs/week	Continuous Internal Assessment : 20 Marks
Tutorial:NIL	Attendance-10 Marks
Total Contact Hours:45 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop basic concept of Software Development in Computer Science

Course Objectives:

1. To provide an idea of using various process models in the software industry according to given circumstances.
2. Analysis of Proper Requirement Specification and importance of its documentation.
3. Emphasize on structured approach to handle software development.
4. Understanding of proper software Testing modules and its implementations
5. Inculcate essential technology and software engineering knowledge and skills essential to build reasonably complex usable and maintainable software iteratively.

Course Content:

Contents (Theory)	Hrs	Marks
UNIT 1: Introduction to Software Engineering	**07**	**08**

- The evolving Role of software & changing nature of software.
- Software Engineering –A layered Technology approach.
- A process framework & software project tracking & control.
- The Capability Maturity Model Integration technique.
- Process patterns, process Assessment, personal & Team Process models & Process Technology Theories.
- Process Models –Waterfall, Incremental, RAD, Prototype, Spiral

UNIT 2: Software Development Activities	**10**	**13**

- Software Engineering core principles, Communication, Planning, Modelling, Construction & Deployment principles.
- Requirements Engineering Tasks, Initiating the requirement process.
- Analysis approaches of software & preparation of Analysis model using Data Modelling Concepts, Object oriented Analysis, Flow oriented model, Class-Based model, Behavioural Model.
- Design approaches of software & preparation of design model using Design concepts, Design model, and pattern based design.

UNIT 3 : Software Testing Basics	**10**	**13**

- Software Testing Fundamentals.
- A Strategic approach to software testing.
- Test Strategies for conventional software, Unit Testing, Integration Testing, Regression testing, smoke testing.
- Validation testing using Alpha & beta testing, system testing using recovery, security, stress & performance testing.
- Black Box & White Box Testing.
- Debugging process strategies.

UNIT 4: Project Management	**10**	**14**

- The management spectrum – The people, The product, the process & the project.
- Project scheduling – Basic concepts, relationship between people & effort, effort distribution, defining a task for the software project, Defining a task network & scheduling of project.
- Risk Management – Reactive Vs Proactive risk strategies, software Risks, Risk Identification, Risk Projection & Risk refinement, monitoring & management.
- Change Management – SCM scenario, SCM repository & process.
- Formal method & clean room software development & management approach.

UNIT 5: Software Quality Management& Estimation	**08**	**12**

- Basic Quality Concepts.
- Software Quality Assurance
- Statistical software quality assurance,
- Six sigma strategy.
- Software Reliability
- The ISO 9000 quality standards
- McCall's quality factors.
- Observations on estimation
- The project Planning process, software scope &
- feasibility, Resources
- Decomposition Techniques
- COCOMO II model & the make / Buy design

Reference Books

1. "Fundamental of Software Engineering", Rajib Mall, PHI
2. "Software Engineering –A Practitioner's Approach*"- Roger S. Pressman, TMH*
3. "Software Engineering" Sommerville , Pearson
4. "An Integrated Approach to Software Engineering"-Pankaj Jalote, Narosa
5. "Software Engineering" Udit Agarwal, S.K.Kataria &Sons
6. "Software Engineering" Sajan Mathew, S. Chand

Course outcomes:

- Students can plan & develop the frame work of project.
- Compare various project process models & use in project planning
- Use the principles of communication, planning, modelling construction & deployment
- Apply testing strategies & methods on software projects.
- Compare various testing methods.
- Identify the duties & responsibilities of People, team leader & stakeholders while planning the software project.
- Schedule the project according to time, size, shape, utility & application
- Monitor & manage the risk during the design of software project.
- Use the parameters of software quality assurance
- Calculate the cost of software, using cost estimation models such as COCOMO II.

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Introduction to Software Engineering	A	4	4	0	08
2.	Software Development Activities	A	4	4	5	13
3.	Software Testing Basics	B	4	4	5	13
4.	Project Management	C	4	4	6	14
5.	Software Quality Management& Estimation	C	4	4	4	12
	Total		**20**	**20**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels (Bloom's revised taxonomy)

Course Title :	**Object Oriented Programming using Java**
Course Code	COPC210
Number of Credits :	3 (L: 3, T: 0, P: 0)
Prerequisites	Any prior knowledge of programming language.
Course code : General	Semester : Fourth
Duration : 15 weeks	Total Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 3 hrs/week	Internal Assessment: 10 Marks
Tutorial: - 0 hr/week	Attendance, Assignment & Quiz : - 10, 5, 5 Marks respectively
Total Contact Hours:	45 Hours
	End Semester Examination : 60 Marks

Aim:

Open source platforms play significant role in the corporate world and are gaining popularity because these are freeware and ease of access. Java is a simple, portable, distributive, robust, secure, dynamic, architecture neutral, object oriented programming language. This technology allows the software designed and developed once for an idealized 'virtual machine' and run on various computing platforms. Companies of all sizes are using Java as the main programming platform to develop various applications/projects worldwide. The aim of this course is that student should learn platform independent object oriented programming and java as base language for advanced technology like three tier architecture applications, cloud computing and web development. Many commercial applications as well as developing mission critical applications are using Java Technologies. This necessitates the corporate sectors to hire highly skilled Java developers. So, after learning this course, student can float themselves as Java developer in the software industry as well this course works as foundationcourse for advance Java programming for the forthcoming semester.

Course Objectives:

- Design and implement classes and methods
- Understand and implement basic programming constructs
- Apply object-oriented features to real time entities
- Understand and implement the concept of reusability and extensibility
- Create packages and interfaces and used it in programs
- Design and implement multithreaded programs
- Manage errors and exceptions
- Create packages and interfaces and used it in programs
- Design and implement applet and graphics programming
- Make use of Data streams in programs
- Make use of Data streams in programs

Course Content:		
Contents (Theory)	Hrs./Unit	Marks
UNIT 1: Introduction to Java	5	7
1.1 Basics of Java, Background/History of Java, Java and the Internet, Advantages of Java 1.2 Java Virtual Machine & Byte Code 1.3 Java Environment Setup 1.4 Java Program Structure 1.5 Procedure-Oriented vs. Object-Oriented Programming concept 1.6 Basics of OOP: Abstraction, Inheritance, Encapsulation, Classes, subclasses and super classes, Polymorphism and Overloading, message communication 1.7 Compiling and running a simple "Hello World" program: Setting Up Your Computer,Writing a Program, Compiling, Interpreting and Running the program, Common Errors		
UNIT 2: Building Blocks of theLanguage	**8**	10
2.1 Primitive Data Types: Integers, FloatingPoint type, Characters, Booleans etc 2.2 User Defined Data Type 2.3 Identifiers & Literals 2.4 Declarations of constants & variables 2.5 Type Conversion and Casting 2.6 Scope of variables & default values ofvariables declared 2.7 Wrapper classes 2.8 Comment Syntax 2.9 Garbage Collection 2.10 Arrays of Primitive Data Types 2.11 Types of Arrays 2.12 Creation, concatenation and conversion of a string, changing case of string, character extraction, String Comparison, String Buffer 2.13 Different Operators: Arithmetic, Bitwise, Rational, Logical,Assignment, Conditional, Ternary, Increment and Decrement, Mathematical Functions 2.14 Decision & Control Statements: Selection Statement (if, if...else, switch),Loops (while, do-while, for), Jump statements (break, continue, return & exit)		
UNIT 3: Object Oriented Programming Concepts	**8**	14
3.1 Defining classes, fields and methods, creating objects, accessing rules, thiskeyword, static keyword, method overloading, final keyword. 3.2 Constructors: Default constructors, Parameterized constructors, Passing object as a parameter, constructor overloading.		
UNIT 4: Inheritance, Packages & Interfaces	**10**	12

4.1 Basics of Inheritance, Types of inheritance: single, multiple, multilevel, hierarchical and hybrid inheritance, concepts of method overriding, extending class, super class, subclass, dynamic method dispatch & Object class 4.2 Creating package, importing package, access rules for packages, class hidingrules in a package. 4.3 Defining interface, inheritance on interfaces, implementing interface,multiple inheritance using interface.		
UNIT 5: Exception Handling & Multithreaded Programming	**10**	12
5.1 Types of errors, exceptions, try catch statement, multiple catch blocks, throwand throws keywords, finally clause, usesof exceptions, user defined exceptions 5.2 Creating thread, extending Thread class, implementing Runnable interface, life cycle of a thread, Thread priority & thread synchronization, exception handing in threads.		
Unit 6: File Handling	**4**	5
6.1 Stream classes, class hierarchy, useful I/Oclasses, creation of text file, reading and writing text files		
Reference Book		
1. Core Java - Vol 1, 11e by Horstmann, Pearson; 2. Java by Anita Seth & B.L. Juneja, OXFORD; 3. Core Java for Beginners: A Simplified Approach (Covers Java SE 13) by Shah and Shah, SPD; 4. Java How to Program: Early Objects, 11e by Deitel, Pearson; 5. Java: The Complete Reference, Herbert Schildt, Tata McGraw Hill;		

Course outcomes:

The theory should be taught in such a manner that studentsare able to acquire different learning out comes in cognitive, psychomotor and affective domain to demonstrate following course outcomes.

i. Explain object oriented programming concepts of java.
ii. Comprehend building blocks of OOPs language, inheritance, package andinterfaces.
iii. Identify exception handling methods.
iv. Develop multithreading object oriented programs.
v. Develop an object oriented program handling data file.

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Introduction to Java	A	4	3	0	7
2.	Building blocks of the Language	A	2	2	6	10

3.	Object Oriented Programming Concepts	A	4	4	6	14
4.	Inheritance, Packages and Interfaces	B	2	4	6	12
5.	Exception Handling, Multithreaded Programming	C	4	4	4	12
6.	File Handling	C	0	3	2	05
	Total		**16**	**20**	**24**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels (Bloom's revised taxonomy)

Course Title: Minor Project	
Course Code	**Project 202**
Number of Credits	**02 (L: 0, T: 0, P: 4)**
Prerequisites	Having fundamental knowledge of Computers and elementary mathematics.
Course Category	PC
Course code: CST	**Semester: Fourth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Practical: 4 hrs/week Total Contact Hours: 44 Hours	Continuous Internal Assessment: 60Marks External Assessment: 40 Marks

Aim of the Course

- To develop technical skill.
- To make use of hardware in developing Software.
- Analysis of different type of case studies.
- Software package development.
- Industrial practices in installation and maintenance of computers and computer networks.
- Fabrication of computers.
- Fault diagnosis and testing of computers.
- Industrial practices in respect of documentation and fabrication.
- A variety of computers and peripherals in assembly organizations.

Course Objectives

- ✓ Work in Groups, Plan the work and coordinate the work.
- ✓ Develop leadership qualities.
- ✓ Develop Innovative ideas.
- ✓ Practically implement the acquired knowledge.
- ✓ Develop basic technical Skills by hands on experience.
- ✓ Write project report.
- ✓ Develop skills to use latest technology in Computer/Information Technology field.
- ✓ Analyze the different types of Case studies.
- ✓ Use effectively oral, written and visual communication
- ✓ Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study.
- ✓ Identify, analyse and solve problems creatively through sustained critical investigation.

Course Content:

Sr. No.	Topics for Practice (Any one)	Skill Area
01	Notes and Password Manager	JAVA, MySQL
02	Scientific Calculator Application	C /C++ / Python/ VB

03	Evaluation of academic performance of Student(S)	PHP / Java, MySQL
04	Interactive Website Development of an Institute / Organization	JAVA Script/ PHP, MySQL
05	Breakout Ball Game	JAVA AWT, Swing, Frame etc.
06	Chatbot Song Recommender	Python API Chatbot
07	1) Creating the file Server of your own : You can access the directory of the machine on which the Server code is running from the client machine using Sockets. Explore 'File' class in Java and you could better do that. You can also download the files. All you have to do is write a code for copying (Copying a file over a network) 2) Removable device detection system in LAN : Well, this is something again very interesting that you can do. The class used is same and a bit of logic. Whenever any hosts in the LAN adds up a directory or connects a USB removable drive, the server gives a notification with the hosts IP-Address 3) LAN Based Examination System	JAVA Networking

Course outcomes

After completing the course:

- To enable students to implement Project Planning in their Industrial In-plant Training Project work.
- To be capable of self-education and clearly understand the value of achieving Perfection in the respective Project work.
- Apply fundamental and disciplinary concepts and methods in ways appropriate to their areas of study.

Reference Books

Name of Authors	Title of the Book	Edition	Name of the publisher
Verhas Peter	Java Projects	2nd	Packt Publishing Limited
Vishal Kumar Shah	Amazing Python Projects for Beginners		Notion Press
A. Adams, C. Campbell, A. Khan	Programming Books Bundle with Project		Code Academy
Horstmann	Core Java - Vol 1	11e	Pearson
Horstmann	Core Java - Vol 2	11e	Pearson

Course Title: Operating Systems Lab	
Course Code	COPC212
Number of Credits :1	2hrs/week
Prerequisites	Knowledge on Programming language like c/c++
Course Category	PC
Course Code: CST	**Semester: FOURTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Laboratory: 2 hrs/week Total Contact Hours: 30 Hours	Marks Continuous Internal Assessment:60 Marks External Assessment: 40

Course Objectives:

This Lab course is intended to practice and do experiment on concepts taught in theory class of 'Operating Systems' and gain insight into functioning of the Operating Systems.

Course Content:

Sr. No.	Topics for Practice	
		Skill Area
1	Revision practice of various commands like man, cp, mv, ln, rm, unlink, mkdir, rmdir, etc and many more that were learnt in IT Workshop course and later.	LINUX
2	Simulate the CPU scheduling algorithms - Round Robin, SJF, FCFS, priority	C/C++
3	Simulate Bankers algorithm for Deadlock Avoidance and Prevention	C/C++
4	Simulate all FIFO Page Replacement Algorithm using C program	C/C++
5	Simulate all LRU Page Replacement Algorithms using C program	C/C++
6	Simulate Paging Technique of Memory Management	C/C++
7	Practice various commands/utilitiessuch as catnl, uniq, tee, pg,comm, cmp, diff, tr, tar, cpio, mount, umount, find, umask, ulimit, sort, grep, egrep,fgrep cut, paste, join, du, df , ps, who, etc and manymore.	Linux

Course outcomes

- Students should be able to demonstrate basic knowledge about Operating System,
- Be able to apply OS concepts such as processes, memory and file systems to systemdesign,
- Be able to configure OS in an efficient and secure manner, and become an advance user of operating system.

Reference Books

- Operating Systems- Design and Implementation, Tanenbaum and Woodhull, PearsonEducation India
- Operating Systems- a concept based approach, Kaur and Gupta, Wiley
- UNIX Concepts and Applications, Sumitabha Das, McGraw-Hill Education

Course Title: Introduction to DBMS Lab	
Course Code	**COPC214**
Number of Credits : 1	**2 (L: 0, T: 0, P: 2)**
Prerequisites	Basic Operations on Computer
Course Category	PC
Course Code: CST	**Semester: Fourth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Laboratory: 2 hrs/week Total Contact Hours: 30 Hours	Continuous Internal Assessment: 60Marks External Assessment: 40 Marks

Course Objectives:

This Lab course is intended to practice whatever is taught in theory class of 'Introduction to DBMS'. A few sample case studies are listed with some suggested activities. More case studies may be addedto this list. You need to develop these case studies, apply all relevant concepts learnt in theory classas the course progress, identify activities/operations that may be performed on the database. It willbe a good idea to also use concepts learnt in the course on Software Engineering/SSAD. Use of any "open-source database software" is highly appreciated.

Course Content:

Sr. No.	**Topics for Practice**	Skill Area
01	• Creating & Executing DDL in SQL. • Creating & Executing Integrity constraints in SQL. • Creating & Executing DML in SQL. • Executing relational, logical and mathematical set operators using SQL. • Executing group functions • Executing string operators & string functions. • Executing Date & Time functions. • Executing Data Conversion functions. • Executing DCL in SQL. • Executing Sequences and synonyms in SQL.	Familiarization with simple queries in SQL.
02	• Program for declaring and using variables and constant using PL/SQL. • Program using if then else in PL/SQL. • Program using for loop & while loop in PL/SQL. • Program using nested loop in Pl/SQL.	Familiarization with PL/SQL.

03	Employee database – 'Create' employee table, 'Select' and display an employee matching a given condition, 'Delete' duplicate records, delete rows using triggers,insert and update records, find net salary, etc.	Sample case Studies
04	Students' Academic database consisting of relevant tables to handle simple queries with PL/SQL.	Sample case Studies
05	Inventory Management System database. Do related activities such as prepare E-R Model, Relational Model, do Normalization, Create Tables, Insert data, Delete Data, Query database, create stored procedures, etc.	Sample case Studies
06	Bus Operator (Roadways) – Do related activities such as prepare E-R Model, Relational Model, do Normalization, Create Tables, Insert data, Delete Data, Query database, create stored procedures, etc.	Sample case Studies

Course outcomes

After completing the course, the students will understand –

- How to design a database, database-based applications.
- How to use a DBMS in Commercial applications.
- The critical role of database in designing for database software systems or database applications.

Reference Books

Name of Authors	Title of the Book	Edition	Name of the publisher
Joel Murach	Murach's SQL and PL/SQL	2ND	SPD
Allen	Introduction to Relational Databases and SQL programming.		Wiley
Deshpande	SQL and PL/SQL for Oracle 11g		Dreamtech
Dasgupta	Database Management System, Oracle. SQL and PLSQL		PHI
Priyadarsini	Database management System		Scitech

Course Title : Computer Networks Lab	
Course Code	COPC216
Number of Credits :	1 (L:, T: 0, P: 2)
Prerequisites	An understanding of the TCP/IP protocol and the its layer model is recommended. Basic knowledge of python (such as through Intro to Computer Science) is required.
Course Category	CST
	Semester : Fourth
Duration : 15 weeks	Total Marks : 100
Teaching Scheme	**Examination Scheme**
Laboratory: 2 hrs/week **Total Contact Hours: 30 Hours**	Continuous Internal Assessment: 60 Marks External Assessment: 40 Marks

Aim:

The practical should be carried out in such a manner that students are able to acquire different learning out comes in cognitive, psychomotor and affective domain to demonstrate following course outcomes.

i. Describe various protocols, models in networks.
ii. Explain operations of TCP, HTTP, and DNS.
iii. Illustrate use of Subnets, Ipv4 in computer networks.
iv. Design simple computer networks.
v. Establish and Commission simple computer networks
vi. Identify and solve network operational problems.

S.No.	Topics for Practice
1	Showing various types of networking cables and connectors, identifying them clearly
2	Looking at specifications of cables and connectors of various companies on Internet, find out differences.
3	Making patch cords using different types of cables and connectors - crimping, splicing, etc
4	Demonstration of different type of cable testers, using them for testing patch cords pre-pared by the students in Lab and standard cables prepared by professionals
5	Configuring computing devices (PC, Laptop, Mobile, etc) for network, exploring different options and their impact – IP address, gateway, DNS, security options, etc
6	Showing various networking devices – NICs, Hub, Switch, Router, WiFi access point, etc.
7	Looking at specifications of various networking devices various companies on Internet, find out differences.
8	Network simulation tool (e.g. Cisco Packet Tracer)
9	Setting up a small wired LAN in the Lab
10	Setting up a small wireless LAN in the Lab

SUGGESTED LIST OF STUDENT ACTIVITIES

i. Identify type of Network in your Institute.
ii. Prepare a design of Network in your Institute
iii. Visit your Institute server room and various places where Racks and servers installed, identify various Network components, collect information about installation of necessary hardware and software.
iv. Visit any ISP in your area.
v. Prepare Charts of Network Topologies.
vi. Seminar presentations.

Reference Books:

1. Cisco press books on CCNA
2. User manual of networking devices available in the lab
3. Wiki pages on networking devices

Course outcomes:

1. Understanding of computer networks, issues, limitations, options available.
2. Able to configure basic small LAN and connect computers to it.

List of Major Equipment/ Instrument with Broad Specifications

i. Computer systems
ii. Network Cable Cat 5/Cat 6.
iii. Crimping Tool
iv. UTP Cable Tester
v. Layer 2 Switch
vi. Wireless Access point and Wireless router
vii. Impacting Tool
viii. Network cable connectors
ix. Network Trainer Kit.

Course Title : Object Oriented Programming Lab using Java	
Course Code	COPC218
Number of Credits :	1 (L:, T: 0, P: 2)
Prerequisites	Any prior knowledge of programming language.
Course Category	CST
Course code : General	Semester : Fourth
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	**Examination Scheme**
Laboratory: 2 hrs/week **Total Contact Hours: 30 Hours**	Continuous Internal Assessment: 60Marks External Assessment: 40 Marks
Aim:	Develop software applications using object oriented concept in a Java SDK environment

Course Objectives:

1.Understand fundamentals of programming such as variables, conditional and iterative execution, methods, etc.
2. Understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
3. Be aware of the important topics and principles of software development.
4. Have the ability to write a computer program to solve specified problems.
5. Be able to use the Java SDK environment to create, debug and run simple Java programs

The practical/exercises are designed to develop different types of skills of the competency. Following is the list of practical problems.

Sr. No	**Unit No.**	**Exercise/Practical** (Outcomes in Psychomotor Domain)
1	1	Install JDK, write a simple “Hello World” or similar java program, compilation, debugging, executing using java compiler and interpreter.
2	2	Write a program in Java to generate first n prime numbers.
3	2	Write a program in Java to find maximum of three numbers using conditional operator
4	2	Write a program in Java to find second maximum of n numbers without using arrays
5	2	Write a program in Java to reverse the digits of a number using while loop
6	2	Write a program in Java to convert number into words & print it
7	2	Write programs in Java to use Wrapper class of each primitive data types
8	2	Write a program in Java to multiply two matrix

9	3	Write a static block which will be executed before main( ) method in a class.
10	3	Write a program in Java to demonstrate use of **this** keyword. Check whether **this** can access the private members of the class or not.
11	3	Write a program in Java to develop overloaded constructor. Alsodevelop the copy constructor to create a new object with the state of the existing object.
12	3	Write a program in Java to demonstrate the use of private constructorand also write a method which will count the number of instances created using default constructor only.
13	3	Write a program in Java to demonstrate the use of **'final'** keyword in the field declaration. How it is accessed using the objects.
14	3	Develop minimum 4 program based on variation in methods i.e. passing by value, passing by reference, returning values and returning objects from methods.
15	4	Write a program in Java to demonstrate single inheritance, multilevel inheritance and hierarchical inheritance.
16	4	Create a class to find out whether the given year is leap year or not. (Use inheritance for this program)
17	4	Write an application that illustrates how to access a hidden variable. Class **A** declares a static variable **x**. The class **B** extends **A** and declaresan instance variable **x**. **display( )** method in **B** displays both of these variables.
18	4	Write a program in Java in which a subclass constructor invokes the constructor of the super class and instantiate the values.
19	4	Write a program that illustrates interface inheritance. Interface **P12** inherits from both **P1** and **P2**. Each interface declares one constant and one method. The class **Q** implements **P12**. Instantiate **Q** and invoke each of its methods. Each method displays one of the constants.
20	4	Write an application that illustrates method overriding in the same package and different packages. Also demonstrate accessibility rules in inside and outside packages.
21	4	Describe **abstract** class called **Shape** which has three subclasses say **Triangle**, **Rectangle**, **Circle**. Define one method **area()**in the abstract class and override this **area()** in these three subclasses to calculate for specific object i.e. area() of Triangle subclass should calculate area of triangle etc. Same for **Rectangle** and **Circle**
22	4	Write a program in Java to demonstrate implementation of multiple inheritance using interfaces.
23	4	Write a program in Java to demonstrate use of final class.
24	5	Write a program in Java to develop user defined exception for 'Divide by Zero' error.
25	5	Write a program in Java to demonstrate multiple try block and multiple catch exception
26	5	Write an small application in Java to develop Banking Application in which user deposits the amount Rs 1000.00 and then start withdrawingof Rs 400.00, Rs 300.00 and it throws exception "Not Sufficient Fund" when user withdraws Rs. 500 thereafter.

27	5	Write a program that executes two threads. One thread displays "Thread1" every 2,000 milliseconds, and the other displays "Thread2" every 4,000 milliseconds. Create the threads by extending the **Thread** class
28	5	Write a program that executes two threads. One thread will print the even numbers and the another thread will print odd numbers from 1 to50.
29	5	Write a program in Java to demonstrate use of synchronization of threads when multiple threads are trying to update common variable.
30	6	Write a program in Java to create, write, modify, read operations on a Text file.

SUGGESTED LIST OF STUDENT ACTIVITIES

Following is the list of proposed student activities like:

i. Study available small Java application on internet and reuse in your application
ii. Develop Java object oriented application programs
iii. Present the application developed

Reference Book:

1. Practical Java Project for Beginners, B. M. Harwani, SPD;
2. Intro to Java Programming, Comprehensive Version, 10e, Liang, Pearson;

Course Outcome

On completion of the course the student should be able to:

- Use an integrated development environment to write, compile, run, and test simple object-oriented Java programs.
- Read and make elementary modifications to Java programs that solve real-world problems.
- Validate input in a Java program.
- Identify and fix defects and common security issues in code.
- Document a Java program using Javadoc.
- Use a version control system to track source code in a project.

(A) List of Major Equipment/Materials

i. Computer System with latest configuration and memory
ii. Multimedia projector
iii. Internet Access
iv. Access to library resources

(B) List of Software/Learning Websites

i. Java Development Kit: http://www.oracle.com/technetwork/java/javase/downloads/index.html
ii. http://docs.oracle.com/javase/specs/jls/se7/html/index.html
iii. http://docs.oracle.com/javase/tutorial/java/index.html
iv. http://www.tutorialspoint.com/java/
v. http://www.learnjavaonline.org/

vi. http://www.c4learn.com/javaprogramming/
vii. http://www.learn-java-tutorial.com/
viii. http://www.tutorialspoint.com/javaexamples/

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Computer Science & Engineering [CSE], Computer Science & Technology [CST], Computer Software Technology [CSWT] & Information Technology [IT]

Part-III (5th Semester)

Revised 2022

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Semester V

Sl. No	Category	Code No.	Course Title	Hours per wee			Total contact hrs/ week	Credits
				L	T	P		
1.	Program core course	COPC301	Microprocessor & Microcontroller (based on 8086 & 8051)	3	0	0	3	3
2.	Program core course	COPC303	IoT	3	1	0	3	4
3.	Program Elective Course-1	COPE304 / ***	Program Elective-1 (any one) i) Mobile Computing ii) Advanced Computer Network	3	1	0	4	4
4.	Program Elective course-2	COPE305 / ***	Program Elective-2 (any one) i) Theory of Automata ii) Fundamentals of AI	3	1	0	4	4
5.	Program Elective course-3	COPE306/ ***	Program Elective-3 (any one) 1) Computer Graphics 2) Digital Image Processing	3	1	0	4	4
6.	Program core course		Microprocessor & Microcontroller Lab using simulator/debug	0	0	2	2	1
7.	Summer Internship-II (4 to 6 weeks)	SI301	Summer Intern-ship-2					1
8.	Major Project	PR302		0	0	4	4	^
	Total Credits							**21**

*** Will be mentioned by the subject name.

^1 credit is carried forward from the Vth semester major project evaluation.

Course Title: MICROPROCESSOR & MICROCONTROLLER	
Course Code	COPC301
Number of Credits :2	3 (L: 3, T: 0, P: 0)
Prerequisites	Basic knowledge of Digital Electronics, Computer Organization
Course Category	PC
Course code: CST	**Semester: FIFTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs/week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc: 10 Marks End Semester Examination: 60 Marks

Course Objectives

- **Students will learn working principle of microprocessor and microcontroller and the basic differences among them.**
- **To understand the internal architecture of 8086 and 8051.**
- **To understand the addressing modes and instruction set of 8086 and 8051.**
- **To understand assembly language.**
- **To learn how to write down the programs by using instructions.**
- **To learn the interfacing IO and other devices with 8086**

Course Content:

Contents (Theory)	Hrs./Unit	Module
UNIT 1: Introduction to Microprocessor Architecture of 8086	10	A

- **Introduction and evolution of Microprocessors**
- **Internal Architecture of 8086 and pin diagram**
- **Register Organization of 8086**
- **Introduction to 80286, 80386, 80486 and Pentium (brief description about architectural advancements only).**
- **Memory Organization and Segmentation of 8086**
- **Minimum and Maximum mode operations of 8086**
- **8086 Control signal interfacing , Read and write cycle timing diagrams.**

**** Some portions has already been covered in 4th sem.**

UNIT 2: Programming and Interrupt structure	**6**	A

- **Instruction sets of 8086**
- **Addressing modes**
- **Assembly language and Assembler directives**
- **Interrupts of 8086.**

UNIT 3: Interfacing with 8086	12	B

- **8255 PPI:**
  - **Internal Architecture of 8255**
  - **Modes of operation in details with example.**
  - **Interfacing A to D converters and Interfacing D to A converters (just concept)**
  - **Stepper motor interfacing**
  - **Architecture and interfacing of 8251 USART**
  - **Architecture and interfacing of 8254 Timer/counter**
  - **Architecture and interfacing of DMA controller (8257)**
  - **Architecture 8259 Programmable Interrupt Controller (8259), Command words and operating modes of 8259**

UNIT 4: Introduction to 8051 Microcontroller	10	C

- **Basic difference between Micro-processor and Microcontroller**
- **Overview of 8051 Microcontroller**
- **Internal Architecture of 8051**
- **Memory Organization in details (RAM and ROM).**
- **SFRs of 8051**

UNIT 5: 8051 interfacing and programming	7	C

- **I/O ports and Interrupts**
- **Timers and Counters**
- **Serial Communication**
- **Instruction set and Addressing modes.**

Course outcomes

After the successful completion of the course the student should be able to:

- **understand the Microprocessor capability in general and explore the evaluation of microprocessors.**
- **understand the addressing modes of Microprocessors**
- **understand the Microcontroller capability and its usage.**
- **write down programs of Microprocessors and Microcontrollers.**
- **interface Microprocessors and Microcontrollers with other electronic devices**
- **understand the interrupt system of both.**

Reference Books

1. Microprocessors and Interfacing, Douglas V Hall, Mc–Graw Hill, 2nd Edition.
2. Ray and Burchandi, “Advanced Microprocessors and Interfacing”, Tata McGraw–Hill.
3. Kenneth J Ayala, “The 8051 Microcontroller Architecture, Programming and Applications”, Thomson Publishers, 2nd Edition.
4. PIC Microcontroller and Embedded Systems using Assembly and C for PIC 18, - Muhammad Ali Mazidi, RolindD.Mckinay , Danny causey -Pearson Publisher
5. Microprocessor ans microcontroller, Krishna Kant, PHI

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Course Title :	**Internet of Things**
Course Code	COPC303
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : FIFTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 4 hrs/week	Continuous Internal Assessment : 20 Marks
Lectures:- 3hrs/week **Tutorial: 1 hr/week**	Attendance-10 Marks
Total Contact Hours:60 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop basic concepts of IoT and its applications

Course Objectives:

The Internet is evolving to connect people to physical things and also physical things to other physical things all in real time. It's becoming the Internet of Things (IoT). The course enables student to understand the basics of Internet of things and protocols. It introduces some of the application areas where Internet of Things can be applied. Students will learn about the middleware for Internet of Things. To understand the concepts of Web of Things.

Course Content:

Contents (Theory)	Hrs	Marks	Module
UNIT 1: IoT	**10**	10	A

- **What is the IoT and why is it important? Elements of an IoT ecosystem,**
- **Technology drivers, Business drivers,**
- **Trends and implications,**
- **Overview of Governance,**
- **Privacy and Security Issues.**
- **Sensing**
- **Actuation**

UNIT 2: IoT PROTOCOLS	**14**	**14**	A

Protocol Standardization for IoT – Efforts – M2M and WSN Protocols – SCADA and RFIDProtocols – Issues with IoT Standardization – Unified Data Standards – Protocols – IEEE802.15.4–BACNet Protocol– Modbus – KNX – Zigbee– Network layer – APS layer – Security

UNIT 3: IOT ARCHITECTURE	**10**	**10**	B
IoT Open source architecture (OIC)- OIC Architecture & Design principles- IoT Devices and deployment models- IoTivity : An Open source IoT stack - Overview- IoTivity stack architecture- Resource model and Abstraction.			
UNIT 4: WEB OF THINGS	**10**	**10**	B
Web of Things versus Internet of Things – Two Pillars of the Web – Architecture Standardization for WoT– Platform Middleware for WoT – Unified Multitier WoT Architecture – WoT Portals and Business Intelligence.			
UNIT 5: Arduino	**6**	**6**	C
Introduction to Arduino programming, Integration of Sensors/Actuators to Arduino			
UNIT 6: IoT APPLICATIONS	**10**	**10**	C
IoT applications for industry: Future Factory Concepts, Brownfield IoT, Smart Objects, Smart Applications. Study of existing IoT platforms /middleware, IoT- A, Hydra etc.			

Reference Books:

1. IoT Fundamentals, 1e, Hanes, Pearson Education
2. Enterprise IoT, Slama, SPD
3. Internet of Things, Ramgir, Pearson Education
4. Honbo Zhou, "The Internet of Things in the Cloud: A Middleware Perspective", CRC Press,2012.
5. Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), "Architecting the Internet ofThings", Springer, 2011.
6. https://nptel.ac.in/noc/individual_course.php?id=noc17-cs22

Course outcomes:

Students will have good understanding of various aspect of IoT, know some tools and have basic implementation skills

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Unit 1	A	4	4	2	10
2.	Unit 2	A	4	4	6	14
3.	Unit 3	B	4	2	4	10
4.	Unit 4	B	6	2	2	10
5.	Unit 5	C	2	2	2	06
6.	Unit 5	C	2	4	4	10
	Total		**22**	**18**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels(Bloom's revised taxonomy)

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)

Course Title :	MOBILE COMPUTING
Course Code	COPE304/1
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : FIFTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 4 hrs/week	Continuous Internal Assessment : 20 Marks
Lectures:-3hrs/week **Tutorial: 1 hr/week**	Attendance-10 Marks
Total Contact Hours:60 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop basic concept of Mobile Computing and its applications

Course Objectives:

1. **To acquire some basic knowledge of Mobile Communications**
2. **To get exposed to Mobile computing and Wireless Networks.**
3. **To learn concepts of Mobile IP and Mobile Ad-Hoc Networks.**
4. **To study the Mobile Computing Operating systems and Mobile Computing Application Environment.**
5. **To understand the relation between Mobile Computing and E-commerce**

Course Content:

Contents (Theory)	Hrs	Marks	Module
UNIT 1: Basics of Mobile Communications	**10**	10	A

- **Mobile handsets**
- **Wireless Communications and Server Applications**
- **Cell phone System**
- **Types of Telecommunication Networks**
- **Components of wireless communication system**
- **Architecture of mobile telecommunication system**
- **Wireless networking standards**
- **Wireless LANs and Wireless LAN Architecture**
- **Applications of WLANs and Advantages of WLANs over wired LANs**
- **Bluetooth Technology and Protocol stack of Bluetooth.**

UNIT 2: Mobile Computing and Wireless Network	**12**	**12**	**A**

- **Concept of Mobile Computing**
- **Comparison of Mobile Computing and Wireless Networking**
- **Mobile Computing Application**
- **Characteristics of Mobile Computing**
- **Structure of Mobile Computing Application**
- **Cellular Mobile Communication**
- **Generation of Cellular Communication Technologies**
- **Global System for Mobile communications(GSM),GSM Services, System Architecture of GSM,GSM security**
- **General Packet Radio Service(GPRS),GPRS Services, GPRS Architecture**
- **Universal Mobile Telecommunication System (UMTS),UMTS Network Architecture**
- **Software Defined Radio(SDR)**
- **Mobile phone and human body.**

UNIT 3 : Mobile IP and Mobile Ad Hoc Networks	**18**	**18**	**B**

- ✓ **Mobile IP and Packet Delivery**
- ✓ **Desirable features of Mobile IP**
- ✓ **Key mechanism used in Mobile IP and Route Optimization**
- ✓ **Dynamic Host Configuration Protocol(DHCP)**
- ✓ **Significance of DHCP**
- ✓ **Basics concepts of Ad Hoc Network setup**
- ✓ **Characteristics of Mobile Ad Hoc Networks(MANETs) and its Operational Constraints and design issues**
- ✓ **Applications of MANETs**
- ✓ **Proactive Routing protocol-DSDV**
- ✓ **Reactive Routing Protocols — DSR, AODV**
- ✓ **Hybrid routing –ZRP**
- ✓ **Multicast Routing- ODMRP**
- ✓ **Vehicular Ad Hoc Networks(VANETs), Difference of MANET and VANET**
- ✓ **Security issues in a MANET**

UNIT 4: Mobile Computing Operating systems and Mobile Computing Application Environment	**12**	**12**	**C**

- ✓ **Mobile Device Operating Systems**
- ✓ **Special Constraints & Requirements**
- ✓ **Commercial Mobile Operating Systems**
  - **Palm OS**
  - **Symbian OS**
  - **iOS**
  - **Android**
  - **BlackBerry**
  - **Windows Phone**
- ✓ **Mobile Devices as Web Clients ,**
- ✓ **HDML(Handheld Mark-up Language) ,WAP, J2ME - J2ME Configuration,**
- ✓ **Android Application Development - Software Development Kit(SDK), Features of SDK,**
- ✓ **Android Application Components, Android Software stack Structure, Advantages of Android.**

UNIT 5: Mobile Computing and E-commerce	08	08	C

- **Application of M-Commerce**
- **Business to Consumer(B2C) Applications**
- **Business to Business (B2B) Applications**
- **Structure of M-Commerce, Pros and Cons of M-Commerce**
- **Mobile Payment System and Mobile Payment Schemes**
- **Desirable properties of a Mobile Payment system**
- **Mobile Payment solutions**
- **Process of Mobile Payment**
- **Security Issues**

Reference Books:

1.Fundamentals of Mobile Computing, Prasant Kumar Pattanaik, Rajib Mall, PHI

2. Mobile Computing :Technology Applications and Service Creation, Ashok K Talukdar Hasan Ahmed & Roopa R Yavagal, McGrawHill

3. Mobile Computing, Raj Kamal, OXFORD

Course outcomes:

- **Recognize and explain wireless and Mobile Communication system and Bluetooth technology.**
- **Can describe and differentiate Mobile Computing vs Wireless Networking, GSM, GPRS, UMTS and SDR.**
- **Can explain the working of Mobile IP and Mobile Ad Hoc Networks, Vehicular Ad Hoc Network.**
- **Describe the constraints and survey of commercial mobile Operating Systems.**
- **Discuss and explain Mobile Application Development.**
- **Acquired the knowledge of different Mobile E-Commerce applications.**

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Basics of Mobile Communications	A	3	4	3	10
2.	Mobile Computing and Wireless Network	A	4	4	4	12
3.	Mobile IP and Mobile Ad Hoc Networks	B	5	5	8	18
4.	Mobile Computing Operating systems and Mobile Computing Application Environment	C	6	3	3	12
5.	Mobile Computing and E-commerce	C	2	4	2	08
	Total		**20**	**20**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels(Bloom's revised taxonomy)

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Course Title :	**Advance Computer Networks**
Course Code	COPE304/2
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : FIFTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 4 hrs/week	Continuous Internal Assessment : 20 Marks
Lectures:-3hrs/week **Tutorial: 1 hr/week**	Attendance-10 Marks
Total Contact Hours:60 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop Advance Networking Concepts and its applications

Course Objectives:

Introduce Advance Networking Concepts, Theories and Tools

Course Content:

Contents (Theory)	Hrs	Marks	Module
UNIT 1:	**14**	14	A

- **Review of Networking Basics;**
- **Wireless LAN - Introduction to wireless LANs, IEEE 802.11 WAN-Architecture and Services, Physical Layer - MAC sublayer - MAC management sublayer - Other IEEE 802.11 standards**
- **Advance Topics in IPv4 – Sub-netting, Multicasting, Multicast Routing Protocols (IGMP, PIM, DVMRP);**
- **Advance Topics in TCP – flow management, congestion avoidance, protocol spoofing; IPv6**

UNIT 2:	**10**	**10**	A

- **Telecom Networks, Switching Techniques; Introduction to Frame Relay, ATM, MPLS;**
- **VSAT Communication – Star and Mesh architectures, bandwidth reservation;**
- **Wireless Networks – WiFi, WiMax, Cellular Phone Technologies – GSM, CDMA, 3G, 4G,5G**
- **DHCP-Outing in the Internet-MOSTF DVMRP,**

UNIT 3:	**10**	**10**	B

- **Storage Area Networks- Introduction to Storage Technology, Storage System Architecture**
- **Introduction to Networked Storage, Direct-Attached Storage, SCSI, NAS, IP SAN,**
- **Information Availability & Monitoring & Managing Datacentre,**
- **Securing Storage and Storage Virtualization**

UNIT 4:	**10**	**10**	B

Introduction to Network Security –

- **VLAN**
- **VPN**
- **Firewall**
- **IPS**
- **Proxy Servers**
- **Network Redundancy,**
- **Load Balancers,**
- **Caching, Storage Networks**

QoS (Quality of Service)
Network Monitoring – SNMP,RMON

UNIT 5:	**6**	**6**	C

- **Traffic Engineering Planning, WAP-WAP architecture-WAE-WTA Framework-WAP push services- WAP protocol stack, NEST Cellular Network**
- **Tuning RED for Web Traffic - Introduction, Background Work, Experimental Methods, Experimental Network**

UNIT 6:	**10**	**10**	C

Streaming protocols

Real-Time Transport Protocol **(RTP)**
Real Time Streaming Protocol **(RTSP)**
TCP Friendly Rate Control (TFRC)
Internet Route Access Protocol(RAP)
Session Initiation Protocol**(SIP)**

Reference Books:

1. Internetworking with TCP/IP - *Vol : I, II and III*, Comer / Stevens, Pearson Education
2. An Engineering Approach to Computer Networking, Keshav, Pearson Education
3. Data Communication and Networking, Forouzan, McGrawHill
4. Networking, Anderson & Benedetti, SPD
5. RFCs and Standards Documents (www.ietf.org and other standard body websites)

Course outcomes:

- Analyse wireless LAN technologies including IEEE 802.11.
- Understand internet traffic and plan traffic engineering including IP over ATM and multimedia over internet.
- Design of routing and transport layer protocols for advanced multi hop networks.
- Understanding of cryptographic algorithms for Enterprise networks.
- Understanding of Streaming protocols

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Unit 1	A	4	4	6	14
2.	Unit 2	A	4	4	2	10
3.	Unit 3	B	4	2	4	10
4.	Unit 4	B	6	2	2	10
5.	Unit 5	C	2	2	2	06
6.	Unit 5	C	2	4	4	10
	Total		**22**	**18**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels(Bloom's revised taxonomy)

Course Title: THEORY OF AUTOMATA	
Course Code	**COPE305/1**
Number of Credits : 4 (L: 3, T: 1, P: 0)	
Prerequisites	**Having fundamental knowledge of Computers and elementary mathematics.**
Course Category	PC
Course code: CST	**Semester: FIFTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 4 hrs/week Total Contact Hours: 60 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

This course focuses on the basic theory of Computer Science and formal methods of computation like automata theory, formal languages, grammars and Turing Machines.

Course Objectives

The objective of this course is to explore the theoretical foundations of computer science from the perspective of formal languages and classify machines by their power to recognize languages.

Course Content:

Contents (Theory)	**Hrs./Unit**	**Marks**
UNIT 1: Introduction to Automata	15	10
1.1. Basic idea on Alphabets, Strings and Languages, Automata and Grammars, Regular Languages, Deterministic finite Automata (DFA)-Formal Definition, Simplified notation, State transition graph, Transition table, Language of DFA. 1.2. Nondeterministic finite Automata (NFA), NFA with epsilon transition, Language of NFA, Equivalence of NFA and DFA. 1.3. Minimization of Finite Automata, Distinguishing one string from other. 1.4. FA with output - Moore and Mealy machine, Equivalence of Moore and Mealy Machine, Applications and Limitation of FA.		
UNIT 2: Regular Expression (RE)	15	15
2.1. Definition, Operators of regular expression and their precedence, Algebraic laws for Regular expressions, Kleen's Theorem. 2.2. Arden's Theorem and applications, Conversion of Non-deterministic systems to deterministic system (application), Construction of finite automata equivalent to a regular expression (with application), Equivalence of two finite automata (application), Equivalence of two regular expressions. 2.3. Pumping Lemma for regular Languages, Application of Pumping Lemma, Closure		

properties of Regular Languages.		
UNIT 3: Grammar Formalism	10	15
3.1. Regular grammars - Right linear and left linear grammars, Equivalence between regular linear grammar and FA. 3.2. Context Free Grammar, Definition, Examples, Derivation, Derivation trees, Ambiguity in Grammar, Inherent ambiguity, Ambiguous to Unambiguous CFG. 3.3. Useless symbols, Simplification of CFGs; 3.4. Normal forms for CFGs - CNF and GNF, Closure properties of CFLs; Decision Properties of CFLs-Emptiness, Finiteness and Membership.		
UNIT 4: Push Down Automata (PDA):	10	10
4.1. Description and definition, Instantaneous Description, Language acceptance of PDA. 4.2. Acceptance by Final state, Acceptance by empty stack. 4.3. Deterministic PDA, Equivalence of PDA and CFG, CFG to PDA.		
UNIT 5: Turing Machines (TM)	10	10
5.1. Basic model, Definition and representation, Instantaneous Description, Language acceptance by TM. 5.2. Computable functions, Types of Turing machines, Universal TM. 5.3. Halting problem, Introduction to Undecidability, Undecidable problems about TMs.		

Course outcomes

Student should be able to

- **Understand the basic properties of formal languages and grammars.**
- **Differentiate regular, context-free and recursively enumerable languages.**
- **Make grammars to produce strings from a specific language.**
- **Acquire concepts relating to the theory of computation and computational models including decidability and intractability.**

Reference Books

Name of Authors	Title of the Book	Edition	Name of the publisher
Hopcroft	Introduction to Automata Theory, Languages, and Computation.		Pearson
Mishra & Chandrasekaran	Theory of Computer Science (Automata, Languages and Computation).		Pearson

Kulkarni	Theory of Computation		Oxford
Nagpal	Formal Language and Automata Theory		Oxford
Kandar	Introduction to Automata Theory, Formal Languages and Computation		Pearson

Course Title: Fundamentals of AI	
Course Code	**COPE 305/2**
Number of Credits : 4 (L: 3, T: 1, P: 0)	
Prerequisites	**Having fundamental knowledge of Computers and elementary mathematics.**
Course Category	PC
Course code: CST	**Semester: FIFTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 4 hrs/week Total Contact Hours: 60 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

Artificial Intelligence, or AI, is a unique branch of computer science which aims solely at **creating intelligent machines**. In the last few years, Artificial Intelligence has become a highly important part of the technology industry. Many students are now willing to study Artificial Intelligence and make a career in this particular field.

Course Objectives

- To understand the various characteristics of intelligent agents.
- To learn the different search strategies in AI.
- To learn to represent knowledge in solving AI problems.
- To understand the different ways of designing software agents.
- To know about the various applications of AI.

Course Content:

Contents (Theory)	**Hrs./Unit**	**Marks**
UNIT 1: Introduction to Artificial Intelligence	10	10
1.1. Introduction to AI, Future of Artificial Intelligence. 1.2. Agents and Environment, Concept of rationality, Nature of environment. 1.3. Structure of Agents, Problem solving agents to typical AI problems.		
UNIT 2: PROBLEM SOLVING	10	10
2.1. Search algorithms, informed & uninformed search strategies, Heuristic search strategies, heuristic functions. 2.2. Local search and optimization problems, local search in continuous space, search with non- deterministic actions, search in partially observable environments. 2.3. Online search agents and unknown environments.		
UNIT 3: GAME PLAYING AND CSP	10	15

3.1. Game theory, optimal decisions in games, alpha-beta search. 3.2. Stochastic games, partially observable games. 3.3. Constraint satisfaction problems (CSP), Structure of CSP, constraint propagation, backtracking search for CSP, local search for CSP.		
UNIT 4: LOGICAL AGENTS	15	10
4.1. Knowledge-based agents, propositional logic, Statement and proving of propositional theorem. 4.2. Propositional model checking. 4.3. Agents based on propositional logic, First-order logic, syntax and semantics, knowledge representation and engineering. 4.4. Inferencesin first-order logic, forward chaining, backward chaining, Resolutions.		
UNIT 5: KNOWLEDGE REPRESENTATION AND PLANNING.	15	15
5.1. Ontological engineering, categories and objects, Events. 5.2. Mental objects and modal logic, reasoning systems for categories, reasoning with default information. 5.3. Classical planning, algorithms for classical planning, heuristics for planning, hierarchical planning. 5.4. Non-deterministic domains, time, schedule, and resources, analysis.		

Course outcomes

Student should be able to

- **Use appropriate search algorithms for any AI problem.**
- **Represent a problem using first order and predicate logic.**
- **Provide the apt agent strategy to solve a given problem.**
- **Design software agents to solve a problem.**
- **Design applications for NLP that use Artificial Intelligence.**

Reference Books

Name of Authors	Title of the Book	Edition	Name of the publisher
Stuart Russell and Peter Norvig.	Artificial Intelligence: A Modern Approach.	3rd / 4th	Pearson
Kevin Night, Elaine Rich	Artificial Intelligence		TMH
Deepak Khemani	A First Course in Artificial Intelligence		TMH
Dan W. Patterson	Introduction to AI and ES		Pearson
M.C. Trivedi	A classical approach to Artificial Intelligence		Khanna publishing House
https://nptel.ac.in/courses/106106126/			

Course Title: COMPUTER GRAPHICS	
Course Code	COPE 306/1
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	Basic knowledge of 2D and 3D geometry, Matrix, Programming language
Course Category	PC
Course code: CST	**Semester: FIFTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 4 hrs/week Total Contact Hours: 60 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc: 10 Marks End Semester Examination: 60 Marks

Course Objectives

- **This course prepares students for activities involving the design, development, and testing of modeling, rendering, and animation solutions to a broad variety of problems found in entertainment, sciences, and engineering.**
- **Students will learn:**
- **(1) how to develop interactive programs that use effectively the graphics functionalities available in contemporary personal computers,**
- **(2) the fundamental principles and technologies upon which these functionalities, and possibly their future evolutions, are based, and**
- **(3) the skills for designing and implementing practical graphic solutions to challenging problems in different application domains.**

Course Content:

Contents (Theory)	Hrs./Unit	Module
Unit-1: Basic of Computer Graphics	8	A

Introduction of Coordinate representation and Pixel Graphics output devices: CRT, Raster Scan & Random Scan systems; Color CRT monitors, DVST, flat-panel displays, video controller and raster scan display processor. Graphics Input Devices: Keyboard, Mouse, Track-ball, space ball, Joysticks, data Glove, Light Pen, Digitizer, Image scanners, touch panels, voice systems; Graphics software

Unit-2: Graphics Primitives	**12**	A

Point and Lines, Line Drawing Algorithms: Simple, DDA, Bresenham's Line Drawing algorithm, Midpoint Circle and Ellipse drawing algorithm, Polygon drawing: Representation of polygon; Conventional methods for drawing polygons; Real time Scan Conversion and Run length encoding; Filled area primitives: Scan-Line Polygon Fill Algorithm, Flood-Fill Algorithm, character

generation, Antialiasing		
Unit-3: 2D viewing	15	B
Viewing pipeline, Window-to-viewport transformation, 2-D Clipping, Chen-Sutherland Line Clipping, Liang-Barsky algorithm, Polygon Clipping: Sutherland-Hodgeman and Weiler-Atherton polygon clipping; Text Clipping		
Unit-4: 2D and 3D Transformation	12	C
Scaling, Rotation, Translation, Shearing, Reflection; Homogeneous coordinates, Composite Transformations, Affine transformation; 3-D concepts and representation, Projections: Perspective, Orthographic, Axonometric, Oblique projections		
Unit-5: 3D transformation and viewing	13	C
Curves and surfaces: Spline representations, Bezier curves, B-spline curves , Visible surface detection methods: Back-face detection, depth-buffer, Z- buffer , scan-line method; Illumination models: Basic illumination models: Ambient light, Diffuse reflection, Specular reflection and Phong Model, Warn model, Half-toning and dithering techniques; RGB, YIQ, HSV and CMY color models, Key-frame animation. ****Some C programs should be done to implement different algorithms in tutorial classes.**		
Course outcomes		
After the successful completion of the course the student should be able to: **1. Know and be able to discuss hardware system architecture for computer graphics. This includes, but is not limited to graphics pipeline, frame buffers, and graphic accelerators/co-processors.** **2. Know and be able to design and implement model and viewing transformations,** **3. Know and be able to use the underlying algorithms, mathematical concepts, supporting computer graphics. These include but are not limited to: • Composite 3D homogeneous matrices for translation, rotation, and scaling transformations. • Plane, surface normals, cross and dot products. • Hidden surface detection / removal** **4. Know and be able to select and use among models for lighting/shading.** **5. Know and be able to use and select among current models for surfaces (e.g., geometric; polygonal; hierarchical; mesh; curves, splines).** **6. Be able to discuss future trends in computer graphics and quickly learn future computer graphics concepts and APIs.**		
Reference Books		

1. Computer Graphics C Version, D. Hearn And P. Baker, Pearson Education
2. Computer Graphics, Foley and van Dam, Person Education
3. Computer Graphics with OpenGL, Hearn and Baker, Pearson
4. Procedural Methods for computer graphics, Rogers, TMH
5. Computer Graphics with virtual reality systems, R. K. Maurya, Wiley-India
6. Computer Graphics, Sinha & Udai, TMH

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)

Course Title :	DIGITAL IMAGE PROCESSING
Course Code	COPE306/2
Number of Credits :4	3 (L: 3, T: 1, P: 0)
Prerequisites	NIL
Course Category	PC
Course code : CST	Semester : FIFTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 4 hrs/week	Continuous Internal Assessment : 20 Marks
Lectures:-3hrs/week **Tutorial: 1 hr/week**	Attendance-10 Marks
Total Contact Hours:60 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop basic concept of Image Processing in Computer Science

Course Objectives:

1. **To become familiar with digital image fundamentals**
2. **To get exposed to simple image enhancement techniques in Spatial and Frequency domain.**
3. **To learn concepts of degradation function and restoration techniques.**
4. **To study the color image processing techniques.**
5. **To become familiar with image compression and recognition methods**

Course Content:

Contents (Theory)	Hrs	Marks	Module
UNIT 1: Image Processing Fundamentals	**09**	09	A

- **Overview & Nature of Image Processing**
- **Digital Image Representation & types of Images**
- **Components of Image Processing system**
- **Steps in Image Processing:**
  - ❖ **Elements of Visual Perception**
  - ❖ **Image Sensing and Acquisition**
  - ❖ **Image Sampling and Quantization**
- **Relationships between pixels**
- **2D mathematical preliminaries, 2D transforms - DFT, DCT**
- **Image Processing Applications**

UNIT 2: Image Enhancement in the Spatial Domain	**13**	**13**	**A**

- **Some Histogram Processing in details,**
- **Enhancement Basic Gray Level Transformations,**
- **Using Arithmetic/Logic Operations,**
- **Basics of Spatial Filtering**
- **Smoothing Spatial Filters,**
- **Sharpening Spatial Filters,**
- **Combining Spatial Enhancement Methods**

UNIT 3: Image Restoration.	**13**	**13**	**B**

- **A Model of the Image degradation/Restoration process,**
- **Noise Modelling**
- **Image Restoration in the Presence of Noise Only–Spatial Filtering**
  - ✓ **Arithmetic mean filter**
  - ✓ **Geometric mean filter**
  - ✓ **Median filter**
- **Order Statistics**
- **Adaptive filters**
- **Band reject Filters**
- **Band pass Filters**
- **Notch Filters – Optimum Notch Filtering**
- **Image Restoration Techniques**
  - ➢ **Inverse filter**
  - ➢ **Wiener Filter**
- **Geometric Transformations**

UNIT 4: Color Image Processing	**11**	**11**	**C**

- **Color image storage & processing**
- **Color Models**
  - ➢ **RGB, HSI, HSV, CMY, CMYK color models.**
- **Pseudocolor Image Processing**
- **Basics of Full-Color Image Processing**
- **Color Transformations**
- **Smoothing and Sharpening**

UNIT 5: Image Compression	**14**	**14**	**C**

- **Fundamentals of image compression**
- **Image Compression Models**
- **Compression Algorithms**
- **Error-Free/lossless Compression**
  - ➢ **Run Length Coding**
  - ➢ **Huffman Coding**
  - ➢ **Shannon –Fano Coding**
  - ➢ **Bit-plane Coding**
- **Lossy Compression**
  - ✓ **Lossy Predictive Coding**
  - ✓ **Transform Coding**
- **Image Compression Standards**

- **JPEG standard**
- **MPEG standard**

Reference Books

1. **"Digital Image Processing", Rafael C Gonzalez Richard E. Woods, Pearson**
2. **"Digital Image Processing", Kenneth R. Castleman, Pearson**
3. **"Principles of Digital Image Processing", Wilhelm Burger, Mark J. Burge, Springer**
4. **"Digital Image Processing", S. Sridhar, Oxford**

Course outcomes:

- **Know and understand the basics and fundamentals of digital image processing, such as digitization, sampling, quantization, and 2D-transforms.**
- **Operate on images using the techniques of smoothing, sharpening and enhancement.**
- **Understand the restoration concepts and filtering techniques.**
- **Learn the basics of compression and recognition methods for color models**

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Image Processing Fundamentals	A	5	4	0	09
2.	Image Enhancement in the Spatial Domain	A	3	4	6	13
3.	Image Restoration	B	3	4	6	13
4.	Color Image Processing	C	5	4	2	11
5.	Image Compression	C	4	4	6	14
	Total		**20**	**20**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels(Bloom's revised taxonomy)

Course Title: Microprocessor & Microcontroller Lab using simulator/debug	
Course Code	CST307
Number of Credits :1	2hrs/week
Prerequisites	Knowledge about instruction set and internal architecture of 8086 and 8051
Course Category	PC
Course Code: CST	**Semester: FIFTH**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Laboratory: 2 hrs/week Total Contact Hours: 30 Hours	Continuous Internal Assessment: 60Marks External Assessment: 40 Marks

Course Objectives:

Understanding and implementation of the operation of microprocessors and microcontrollers, machine language programming & interfacing techniques with peripheral devices.

Course Content:

Sr. No.	Topics for Practice	
		Skill Area
1 (8086)	#To Add Two Binary Number Each 1 Bytes Long (without and with carry) #To Add Two Binary Number Each 2 Bytes Long (without and with carry) #To Add array of 1 Bytes numbers. #To Add array of 2 Bytes numbers. #To Add Two Binary Number Each 4/8 Bytes Long #Exchange two memory location without using exchange instruction. #To Find the Maximum Number in a given array. # Short in ascending/descending order Use of DOS interrupt INT 20 and INT 21: # A string is stored in memory location starting from 0200h ended with character $. Display the string. # A one-byte number is stored at 0200h, print the binary of that byte. # Display a hexadecimal byte stored at DL register. # Reading 1/2 digit hexadecimal number from keyboard. ** Writing any other program like using stack, subroutine, code conversion, string manipulation is welcome. # Carry out the interfacing ADC/DAC, Stepper motor etc.	8086

2	# How do mode bits of timer 0 and 1 swapped at the TMOD? # Complement the port P2 bits. # Complement the bit 3 of internal RAM at 21H # Add/Subtract two 8 bit numbers. # Add two 16 bit numbers. # Add two BCD numbers. **Other programs to control timer, subroutine, bit manipulation etc. is welcome. # Use simulator like EDSim51 and perform C programming.	C, Simulator

**** Programs can be written in any mode like by using physical kit, Simulator, Assembler, DEBUG utility etc.**

Course outcomes

On completion of this course, the students will be able to:

- **Program 8086 Microprocessor, 8051 and PIC Microcontrollers for application specific solution**
- **Design microprocessors/microcontrollers-based systems**
- **Implement and develop new experiments on microprocessor/microcontroller based systems.**

Reference Books

1. Microprocessors and Interfacing, Douglas V Hall, Mc–Graw Hill, 2nd Edition.
2. Ray and Burchandi, “Advanced Microprocessors and Interfacing”, Tata McGraw–Hill.
3. Kenneth J Ayala, “The 8051 Microcontroller Architecture, Programming and Applications”, Thomson Publishers, 2nd Edition.
4. PIC Microcontroller and Embedded Systems using Assembly and C for PIC 18, -Muhammad Ali Mazidi, RolindD.Mckinay , Danny causey -Pearson Publisher
5. Microprocessor ans microcontroller, Krishna Kant, PHI

Course Title: Major Project	
Course Code	**PR 302**
Number of Credits	**^^To be assigned in 6th Semester^^**
Prerequisites	Having fundamental knowledge different languages, database, network etc.
Course Category	PC
Course code: CST	**Semester: Fifth**
Duration: 15 weeks	**Marks: ^^ Given in 6th Semester**
Teaching Scheme	**Examination Scheme**
Practical: 4 hrs/week Total Contact Hours: 44 Hours	

Aim of the Course

- To develop technical skill.
- To make use of hardware in developing Software.
- Analysis of different type of case studies.
- Software package development.
- Industrial practices in installation and maintenance of computers and computer networks.
- Fabrication of computers.
- Fault diagnosis and testing of computers.
- Industrial practices in respect of documentation and fabrication.
- A variety of computers and peripherals in assembly organizations.

Course Objectives

- ✓ Work in Groups, Plan the work and coordinate the work.
- ✓ Develop leadership qualities.
- ✓ Develop Innovative ideas.
- ✓ Practically implement the acquired knowledge.
- ✓ Develop basic technical Skills by hands on experience.
- ✓ Write project report.
- ✓ Develop skills to use latest technology in Computer/Information Technology field.
- ✓ Analyze the different types of Case studies.
- ✓ Use effectively oral, written and visual communication
- ✓ Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study.
- ✓ Identify, analyze and solve problems creatively through sustained critical investigation.

Course Content:

Sr. No.	Topics for Practice (Any one)	Skill Area
01	To develop Web pages	HTML,CSS, JAVA Script, MySQL,JSP, ASP
02	To develop Application packages	C/C++/Python/ JAVA/ VB

03	To develop online examination system	PHP / Java, MySQL
04	Develop Web based application	JAVA Script/ PHP, MySQL
05	Android App development	Android studio/Java
06	Image processing application	Python/Mat-lab
	Database oriented application development like Student information system, Library management system etc.	Java/ Python

- Any other project work can be done as guided by project guide (like AI based application, Pattern recognition etc.)
- If someone wants to extend the workflow from the minor project, he or she can be able to extend

Course outcomes

After completing the course:

- To enable students to implement Project Planning in their Industrial In-plant Training Project work.
- To be capable of self-education and clearly understand the value of achieving Perfection in the respective Project work.
- Apply fundamental and disciplinary concepts and methods in ways appropriate to their areas of study.

Reference Books

Name of Authors	Title of the Book	Edition	Name of the publisher
Verhas Peter	Java Projects	2nd	Packt Publishing Limited
Vishal Kumar Shah	Amazing Python Projects for Beginners		Notion Press
A. Adams, C. Campbell, A. Khan	Programming Books Bundle with Project		Code Academy
Horstmann	Core Java - Vol 1	11e	Pearson
Horstmann	Core Java - Vol 2	11e	Pearson

Course Code	**SI 301.**
Course Title	**SUMMER INTERNSHIP II.**
Course Category	**Internship Programme.**
Number of Credits	**1**
Offered to	**4th Semester, Diploma in Comp. Sc. & Tech. Student.**
Pre Requisite	**Elementary knowledge on Computer**

Aim of the Course:

- providing exposure to students in skill development.
- To offer an opportunity for the young students to acquire on job skills, knowledge, attributes and perceptions along with the experience needed to constitute a professional identity beyond Institutional environment.
- To provide means to acquire professional knowledge beyond curriculum.

Course Content:

After the end of **Second Semester**:

- The students are required to involve in Inter/Intra Institutional activities viz. training and simulation program in different Institutes, Polytechnics, Technical Colleges. (OR)
- Even in other Departments within the same Polytechnic. (OR)
- Soft skill training organized by Training & Placement Cell of the respective Institutions; contribution at innovation/entrepreneurship cell of the institute; participation in workshops/competitions etc; Learning at Departmental Lab/ Institutional workshop. (OR)
- Online or Offline Participation in any skilled training related with Computer Science and Technology field for duration of 1 month from any recognized Organization.

Documentation in the form of report should be submitted by the candidates for evaluation purposes.

Course Outcomes

At the end of the course the students will be able to

CO I	To test the theoretical learning in practical situations by accomplishing the task assigned during the internship period.
CO II	Able to apply various soft skills such as time management, positive attitude and communication skills during performance of the tasks assigned in internship organization.
CO III	To assess interest and abilities in the actual field of work.
CO IV	Learn to appreciate time management in real life.

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Computer Science & Engineering [CSE], Computer Science & Engineering [CST], Computer Software Technology [CSWT] & Information Technology [IT]

Part-III (6th Semester)

2023

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Semester VI

Sl. No	Category	Code No.	Course Title	Hours per			Total contact hrs/ week	Credits
				L	T	P		
1.	Program Elective course-4	COPE307/ ***	Program Elective-4 (any one) i) Data Sciences: Data Warehousing & Data Mining, ii) Cloud Computing.	3	1	0	4	4
2.	Humanities and Social Science course	HS302	Entrepreneurship and Start-ups	2	1	0	3	3
3.	Open Elective-1	OE301/* **	Open Elective-1 (Anyone) i)Engineering Economics and Project Management	3	0	0	3	3
4.	Open Elective-2	OE302/***	Open Elective-2 (any one) i) Machine Learning ii) Web Designing	3	0	0	3	3
5.	Major Project	PR302		0	0	6	6	5^
6.	Seminar	SE302		1	0	0	1	1
	Total Credits							**19**

*** Will be mentioned by the subject name.

^2 credit is carried forward from the Vth semester major project evaluation.

Total Credit Point = 82 (Sem 3,4,5,6)

Course Title: Data Warehousing & Data Mining	
Course Code	**COPC207**
Number of Credits: 4 - L: 3, T: 1, P: 0	
Prerequisites	**NIL**
Course Category	PC
Course code: CST	**Semester: Sixth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 4 hrs./week Total Contact Hours: 60 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

This course will introduce the concepts of data ware house and data mining, which gives a complete description about the principles, used, architectures, applications, design and implementation of data mining and data ware housing concepts.

Course Objectives

To introduce the student to various data warehousing and data mining techniques. The course will cover all the issues of KDD process and will illustrate the whole process by examples of practical applications.

✣ To make the student capable of applying data mining techniques in real time applications.

✣ To make the student capable to compare and contrast different conceptions of data mining as evidenced in both research and application.

✣ Explain the role of finding associations in commercial market basket data.

✣ Identify and characterize sources of noise, redundancy, and outliers in presented data.

✣ To get an idea about the data that how it is going to be classified into clusters.

Course Content:

Contents (Theory)	Hrs./Unit	Marks
UNIT 1: DATA WAREHOUSE	15	14

1.1 What Is a Data Warehouse? The need for a Separate Data Warehouse.

1.2 Data Warehouse Models: Enterprise Warehouse, Data Mart and Virtual Warehouse; 1.3 Differences between Operational Database Systems and Data Warehouses, Data Warehouse Modeling: Data Cube, Conceptual Modeling of Data Warehouse.

1.4 Concept Hierarchies, Measures: Their Categorization and Computation.

1.5 OLAP Operations, Operations in the Multidimensional Data Model (OLEP).

1.6 Data Warehouse Design and Usage, From Online Analytical Processing to Multidimensional Data Mining. Data Warehouse Implementation.

UNIT 2: INTRODUCTION TO DATA MINING	10	12
2.1 What is Data Mining? Process of Knowledge Discovery. 2.2 Types of Repositories, Data Mining Functionalities, Methods of presenting Derived Model. 2.3 Data Mining Tasks, Data Mining Trends, Data Mining Issues.		
UNIT 3: ASSOCIATION AND CORRELATION ANALYSIS	8	8
3.1 Basic Concepts, how does Association Rule Learning work? 3.2 The Apriori Algorithm: Basics 3.3 FP Growth Algorithm, Applications of Association Rule Learning.		
UNIT 4: CLUSTERING ALGORITHMS AND CLUSTER ANALYSIS	10	10
4.1 Unsupervised Learning basic idea. 4.2 Clustering Algorithms: K-Means Clustering, K-Medoids clustering (PAM), Hierarchical Clustering, Graph-Based Clustering. 4.3 Cluster Analysis basics, Cluster Evaluation 4.4 Outlier Detection and Analysis		
UNIT 5: CLASSIFICATION	10	8
5.1 Supervised Learning: Classification, Issues regarding Classification, Types of Classifiers: Binary Classification, Multiclass Classification. 5.2 Classification Approaches: Bayesian Classification-Naïve Bayes, Association based Classification, Rule-Based Classifier.		
UNIT 6: WEB MINING	**7**	8
6.1 Web Mining, Mining the web page layout structure. 6.2 Mining web link structure, mining multimedia data on the web. 6.3 Automatic classification of web documents and web usage mining. 6.4 Distributed Data Mining.		

Course outcomes

Student should be able to

Sl. No.	**Description**	**Bloom's Taxonomy Level**
1	**Understand** the functionality of the various data mining and data warehousing component	**Knowledge, Understand**
2	**Appreciate** the strengths and limitations of various data mining and data warehousing models	**Apply, Create**
3	**Explain** the analyzing techniques of various data	**Analyze**
4	**Describe** different methodologies used in data mining and data ware housing.	**Analyze**
5	**Compare** different approaches of data warehousing and data mining with various technologies.	**Evaluating**

Reference Books:

Name of Authors	Title of the Book	Edition	Name of the publisher
Arun K Pujari	Data Mining Techniques	3rd Edition	University Press
David Hand, Heikki Mannila, Padhraic Smyth,	Principles of Data Mining	2012Reprint, Eastern Economy edition	PHI Learning Private Limited
Jiawei Han and Micheline Kamber	Data Mining-Concepts and Techniques	Second Edition	Kaufmann Publishers
Vikaram Pudi, P Radha Krishna	Data Mining	2009	Oxford University Press
Pang-Ning Tan & Michael Steinbach & Vipin Kumar	Introduction to Data Mining,	2nd Edition	Pearson Education

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Course Title :	CLOUD COMPUTING
Course Code	COPE307/2
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	Netwoking Concepts
Course Category	PC
Course code : CST	Semester : SIXTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 4 hrs/week	Continuous Internal Assessment : 20 Marks
Lectures:-3hrs/week **Tutorial: 1 hr/week**	Attendance-10 Marks
Total Contact Hours:60 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	**It will provide the students basic understanding about Cloud Computing, virtualization along with its security aspects and how one can migrate over it.**

Course Objectives:

1. **To learn the fundamental ideas behind Cloud Computing, the evolution of the paradigm, its applicability; benefits, as well as current and future challenges.**
2. **To understand the basics of cloud delivery models.**
3. **To learn about different virtualization techniques that serve in offering software, computation and storage services on the cloud.**
4. **To Analyze the Strategies for Secure Operation the cloud and list of the security requirements**
5. **To comprehend the basic ideas of different cloud tools and applications.**

Course Content:

Contents (Theory)	Hrs	Marks	Module
UNIT 1: Cloud Computing Fundamentals	**11**	11	**A**

- **Origins of Cloud computing. Fundamental concepts and models, Roles and boundaries.**
- **Cloud components.**
- **On-demand self-service, Broad network access, Location independent resource pooling, Rapid elasticity, Measured service.**
- **Comparing cloud providers with traditional IT service providers, Roots of cloud computing**
- **Migrating to clouds.**

UNIT 2: Cloud Delivery Model	**11**	**11**	**A**

- ❖ **Cloud Delivery Models, The SPI Framework.**
- ❖ **Cloud Service Models.**
- ❖ **Cloud Deployment Models.**
- ❖ **Alternative Deployment models**
- ❖ **Expected benefits of the models.**

UNIT 3: Virtualization	**12**	**12**	**B**

- **Characteristics & Taxonomy of virtualization.**
- **Virtualization vs Private Cloud.**
- **Desktop Virtualization, Hardware Virtual Machine (HVM).**
- **Virtual Servers.**
- **Logical Network Perimeter, Network Virtualization**
- **Data Center virtualization, Cloud Storage Device, Cloud usage monitor, Resource replication.**

UNIT 4: Fundamental Cloud Security	**14**	**14**	**B**

- **Cloud Information Security Objectives.**
- **Cloud Security Services & Relevant Cloud Security Design Principles**
- **Secure Cloud Software Requirements.**
- **Secure Development practices, Approaches to Cloud Software Requirement Engineering.**
- **Privacy and Compliance Risks, Threats to Infrastructure,**
- **Data and Access Control, Cloud Service Provider Risks.**
- **Cloud Security Policy Implementation.**

UNIT 5: Cloud Tools and applications	**12**	**12**	**C**

- **Cloud Performance Monitoring tools**
- **General model for Application platform**
- **Apache Virtual Computing Lab, VMWare, CloudSim.**
- **Microsoft Cloud Services (-Azure), Google Cloud Applications, Amazon cloud services.**

Reference Books

1. **"Cloud Computing Concepts, Technology & Architecture"- Thomas Erl, Zaigham Mahmood, and Ricardo Puttini , PrenticeHall**
2. **"Cloud computing a practical approach" - Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill**
3. **"Cloud Computing (Principles and Paradigms)"- Rajkumar Buyya, James Broberg, Andrzej Goscinski, John, Wiley & Sons**
4. **"Cloud Computing"-Shailendra Singh, Oxford**
5. **"Cloud Computing-A Practical approach for learning and Implementation"-A Srinivasan & J. Suresh, Pearson**

Course outcomes:

- **Analyze the Cloud computing setup with its vulnerabilities and applications using different architectures**
- **Apply and design suitable Virtualization concept, Cloud Resource Management**
- **Assess cloud Storage systems and Cloud security, the risks involved, its impact and develop cloud application**
- **Can understand the basics of security service models.**
- **Analyze the Strategies for Secure Operation the cloud architecture and list the security requirements.**

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Cloud Computing Fundamentals	A	4	4	3	11

2.	Cloud Delivery Model	A	4	4	3	11
3.	Virtualization	B	4	4	4	12
4.	Fundamental Cloud Security	B	4	4	6	14
5.	Cloud Tools and applications	C	4	4	4	12
	Total		**20**	**20**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels (Bloom's revised taxonomy)

Course Title: Machine Learning	
Course Code	**OE302**
Number of Credits: 3 - L: 3, T: 0, P: 0	
Prerequisites	**Concept of AI**
Course Category	PC
Course code: CST	**Semester: Sixth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs./week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

This course will introduce the concept of Machine Learning through different learning methods.

Course Objectives

- **To learn the concept of how to learn patterns and concepts from data without being explicitly programmed**
- **To design and analyze various machine learning algorithms and techniques with a modern outlook focusing on recent advances.**
- **Explore supervised and unsupervised learning paradigms of machine learning.**
- **To explore Neural Network and Genetic Algorithm.**

Course Content:

Contents (Theory)	Hrs./Unit	Marks
Unit 1: Supervised Learning (Regression & Classification)	15	20

- ❖ **Basic methods: Distance-based methods, Nearest-Neighbours, Decision Trees, Naive Bayes**
- ❖ **Linear models: Linear Regression, Logistic Regression, Generalized Linear Models**
- ❖ **Introduction to Support Vector Machines, Nonlinearity and Kernel Methods**

Unit 2: Unsupervised Learning	7	10

- **Clustering: K-means/Kernel K-means**
- **Dimensionality Reduction: PCA and kernel PCA**
- **Matrix Factorization and Matrix Completion**

UNIT 3: Artificial Neural Network	8	10

- Neural network representation
- Perception
- Multilayer Network and Back Propagation Algorithm
- Illustrative Example: Face recognition

UNIT 4: Genetic Algorithm	8	10

- Representing Hypotheses
- Genetic Operators
- Fitness Function and Selection
- Hypothesis space search
- Genetic Programming

UNIT 5: Reinforcement Learning	7	10

- **Introduction**
- **The Learning Task**
- **Q Learning**
- **Temporal Difference Learning**

- **Note: Implementation of Machine Learning Algorithm by using suitable software can be done in Project work. Also seminar can be presented on topics of this subject.**

Course outcomes

Student should be able to

Sl. No.	Description	Bloom's Taxonomy Level
1	Understand the concept of machine learning.	**Knowledge, Understand**
2	Identify the regression and classification problem.	Analyze
3	Relate the supervised, unsupervised learning in the real life problem.	**Analyze**
4	Evaluate the machine learning models with respect to the performance parameters.	**Analyze**
5	Design and implement various machine learning algorithms in the range of real world problems.	**Design**

Reference Books:

Name of Authors	Title of the Book	Edition	Name of the publisher
Tom M. Mitchell	Machine Learning	-	Mc Graw Hill
Christopher Bishop	Pattern Recognition and Machine Learning	-	Springer
Rajiv Chopra	Machine Learning	-	Khanna Publishing House
Christopher M. Bishop	Pattern Recognition and Machine	-	Springer

	Learning		

Course Title: Web Designing	
Course Code	OE302/2
Number of Credits: 3 - L: 3, T: 0, P: 0	
Prerequisites	**NIL**
Course Category	PC
Course code: CST	**Semester: Sixth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 3 hrs./week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks

Aim of the Course

This course will introduce the concepts of PHP frameworks, which gives a complete description about the principles, used, architectures, applications, design and implementation of web development concepts. After the completion of course, students will get hands on experience on various techniques of web development and will be able to design and develop a complete website.

Course Objectives

The objective of this course is to provide the necessary knowledge to design and develop dynamic, database-driven web applications using PHP version 5. Students will learn how to connect to any ODBC-compliant database, and perform hands on practice with a MySQL database to create database-driven HTML forms and reports etc. Students also learn how to configure PHP and Apache Web Server. Comprehensive lab exercises provide facilitated hands on practice crucial to develop competence web sites.

Course Content:

Contents (Theory)	Hrs./Unit	Marks	Module
UNIT 1:	7	12	A

- Overview of PHP
- Static vs. Dynamic Web Sites
- Dynamic Content from Databases
- Developing Dynamic Internet Applications
- Client-Side Scripting vs. Server-Side Scripting
- Overview of PHP Advantages and Capabilities
- Configuring PHP.INI
- PHP vs. ASP
- Basic PHP
- echo and print Statements
- Comments in PHP
- PHP Case Sensitivity
- Defining variable and constant
- PHP Data Types
- PHP Operators

➢ Looping Constructs - while, do...while, for, for each			
UNIT 2:	5	10	A
➢ Introduction to the Apache Web Server - What is Apache? - The main benefits / advantages - Apache Installation - Apache Virtual Host - Name-based Virtual Hosts - IP-based Virtual Hosts ➢ PHP Functions - Introduction to Functions - User Defined Functions - Passing Arguments to Functions - Variable scope - Local and Global Scope - Passing Arguments by Reference - Returning Values from a Function - Using Include Files - The Require Statement - Dynamic Function Calls - Recursive Functions - Predefined PHP Functions ➢ PHP Arrays - What is an Array? - Why do we use arrays? - Indexed Arrays - Associative Arrays - Multidimensional Arrays - Sorting Arrays in PHP - Array Functions ➢ PHP MySQL Database and Forms			
UNIT 3:	10	10	B
➢ What is MySQL? - Queries - PHP's Database APIs - Configuring PHP for Database Support - MySQL vs. Access - MySQL vs. SQL Server - Forms and Program - Insert Data Into - Insert Multiple Records Into MySQL - HTTP GET, POST, And Request methods - Insert Data From a Form Into a Database - PHP MySQL Select (Retrieving Data from Forms) - PHP MySQL The Where Clause - PHP MySQL Order By Keyword - PHP MySQL Update - PHP MySQL Delete - Looping through database - PHP Functions Specific to MySQL ➢ Using Cookies with PHP - What is a Cookie? - How to Create a Cookie? - How to Retrieve a Cookie Value? - How to Delete a Cookie? ➢ PHP Sessions - What is a PHP Session - Starting a PHP Session - Storing and Retrieve Session Variable - Session Unset - Destroy A PHP Session ➢ Miscellaneous PHP Tasks - Error Logging - Using Environment Variables - PHP Redirect To Another URL - Getting IP Addresses from Visitors - PHP - Function preg_match()			
UNIT 4:	6	8	B
➢ PHP File Handling - String Functions ➢ E-Commerce Site - What is E-Commerce - E-commerce platforms on the market ➢ SQL Injection - Introduction ➢ PDO - Introduction ➢ Introduction to Frameworks ➢ Introduction to CMS (Content Management System) - WordPress ➢ AJAX ➢ Introduction to open Source CMF (Content management framework) - Codeigniter			
UNIT 5:	17	20	C
➢ Introduction to Codeigniter - What is a PHP Framework - MVC Pattern - Why Should we use a PHP Framework? - When to use a PHP Framework? - What are the Best PHP Frameworks Available? - Codeigniter Overview - CodeIgniter Features - CodeIgniter Basic Concepts and Application Architecture - Installing Codeigniter - Basic Configuration Options - Database Configuration - CodeIgniter - Application Architecture - Directory Structure - CodeIgniter - MVC Framework - Application URL Structure			

- Controller Introduction - Calling a Controller - Creating & Calling Constructor Method - Controller function - Interacting with views
- Views - Views- Introduction - Loading the View - Working with configuration layout - Creating custom layout - Controller variables and parameters - CodeIgniter URLs - Passing argument through url - Redirection - Form and Getting post data
- Models - Model Introduction - Creating Model Class - User defined function in model - Connecting to a Database - Automatic Connecting - Manual Connecting - Inserting Data to Database - Fetching data - Deleting data - Updating data
- Helpers - Helpers – Introduction - Array Helper, Cookie Helper, Date Helper, URL Helper, etc.. - Loading a Helper - Auto load Configuration
- Session Management - Initializing a Session - Add Session Data - To retrieve all session data - To remove all session data - Flashdata - Retrieve Flashdata
- Cookie Management

Course outcomes

Student should be able to

Sl. No.	Description	Bloom's Taxonomy Level
1	**Understand** the functionality of the various PHP syntax	**Knowledge, Understand**
2	**Appreciate** the strengths and limitations of PHP Frame Work	**Apply, Create**
3	**Explain** the analyzing techniques of CodeIgniter	**Analyze**
4	**Describe** different methodologies used in web Designing.	**Analyze**
5	**Compare** different approaches of web designing with various technologies. Develop different type of Web Application in 6th Semester Project work.	**Evaluating**

Note: Development of Web-page can be done as Project work. Also seminar can be presented on topics of this subject.

Reference Books:

Name of Authors	Title of the Book	Edition	Name of the publisher
Ullman	PHP for the Web: Visual Quick-Start Guide	5th Edition	Pearson
Thomas Myer	Professional CodeIgniter		John Wiley & Sons
Welling	PHP and MySQL Web Development	5th Edition	Pearson
Robin Nixon	Learning Php, MySQL & JavaScript: A Step-By-Step Guide to Creating Dynamic Websites	Second 6th Edition	SPD
Ray Harris	Murach's PHP & MySQL		SPD
Michael Morrison, Lynn Beighley	Head First PHP & MySQL		SPD

	A Brain-Friendly Guide		
Dr. Poornima G. Naik, Dr. Girish R. Naik	PHP Coding with CodeIgniter - Hands-on Experience with CodeIgniter		Educreation Publishing

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Unit 1	A	4	6	2	12
2.	Unit 2	A	4	4	2	10
3.	Unit 3	B	4	2	4	10
4.	Unit 4	B	4	2	2	8
5.	Unit 5	C	4	6	10	20
	Total		**20**	**20**	**20**	**60**

Legends: R = Remember; U = Understand; A = Apply and above levels(Bloom's revised taxonomy)

Course Title: Major Project	
Course Code	PR302
Number of Credits: 5	
Prerequisites	**NIL**
Course Category	PC
Course code: CST	**Semester: Sixth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 6hrs./week Total Contact Hours: 90 Hours	Marks: 100

Aim of the Course

Student should able to present their Project work or any other advanced topic. (AI, Cloud computing, Data Mining etc.)

Course Objectives

1. To make them understand the concepts of Project Management for planning to execution of projects.

2. To make them understand the feasibility analysis in Project Management and network analysis tools for cost and time estimation.

3. To enable them to comprehend the fundamentals of Contract Administration, Costing and Budgeting.

4. Make them capable to analyze, apply and appreciate contemporary project management tools and methodologies in Indian context.

Course Content:

This project work is a continuation of 5th Semester Project that can be done individually or in group on topic as described in 5th Semester syllabus.

1.Understand project characteristics and various stages of a project.
2. Understand the conceptual clarity about project organization and feasibility analyses – Market, Technical, Financial and Economic.
3. Analyze the learning and understand techniques for Project planning, scheduling and Execution Control.

Course Title: SEMINAR	
Course Code	SE302
Number of Credits: 1	
Prerequisites	**NIL**
Course Category	PC
Course code: CST	**Semester: Sixth**
Duration: 15 weeks	**Maximum Marks: 100**
Teaching Scheme	**Examination Scheme**
Theory: 1 hrs./week Total Contact Hours: 15 Hours	Marks: 100

Aim of the Course

Student should able to present their Project work or any other advanced topic. (AI, Cloud computing, Data Mining etc.)

Course Objectives

1. Identify and compare technical and practical issues related to the area of course specialization.

2. Outline annotated bibliography of research demonstrating scholarly skills.

3. Prepare a well-organized report employing elements of technical writing and critical thinking.

4. Demonstrate the ability to describe, interpret and analyze technical issues and develop competence in presenting.

Course Content:

1. **Presentation can be done individually or in group**
2. **Presentation can be done on Project work**
3. **Presentation can be done on any advanced topic or emerging fields.**
4. **There should be sufficient number of slides.**
5. **Each student must present their presentation for at least 10 minutes.**
6. **Questions of audience must be answered.**

CO1 - Establish motivation for any topic of interest and develop a thought process for technical presentation.
CO2 - Organize a detailed literature survey and build a document with respect to technical publications.
CO3 - Analysis and comprehension of proof-of-concept and related data.
CO4 - Effective presentation and improve soft skills.